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(54) **USER EQUIPMENT, BASE STATION, AND COMMUNICATION METHOD**

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3GPP TS 38.331 V16.7.0 (Dec. 2021) 3rd Generation Partnership Project; Technical Specification Group Radio Access Network; NR; Radio Resource Control (RRC) protocol specification (Release 16).

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(Continued)

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(57)

ABSTRACT

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H04L 1/1607 (2023.01)

H04W 72/0457 (2023.01)

(52) **U.S. Cl.**

CPC **H04W 72/0446** (2013.01); **H04L 1/1614** (2013.01); **H04W 72/0457** (2023.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

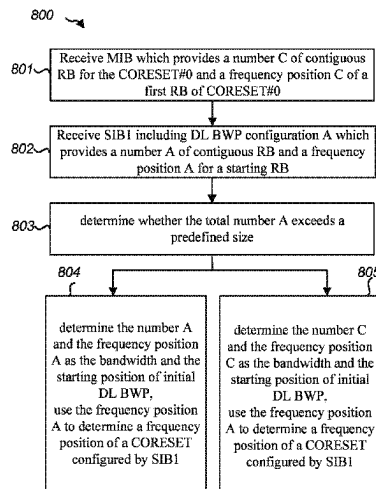
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A method by a user equipment (UE) is described. The method includes receiving, from a base station, MIB providing a first quantity of contiguous resource blocks (RBs) and a first frequency position of a first starting RB for a first CORESET; receiving system information including a second configuration of an initial downlink (DL) BWP, the second configuration providing a second quantity of contiguous RBs, a second frequency position of a second starting RB, and a bitmap indicating frequency domain resources for a second CORESET, each bit of the bitmap corresponds a group of 6 contiguous RBs, with grouping starting from a first RB group in ascending order of RB index; and determining whether the second quantity exceeds a predefined size, wherein in a case the second quantity exceeds the predefined size, determining the first frequency position and the first quantity as a starting position and a bandwidth for an initial DL BWP, and determining a first RB index of the first RB group by using the second frequency position, in a case the second quantity does not exceed the predefined size, determining the second frequency position and the second quantity as the starting position and the bandwidth, and determining the first RB index of the first RB group by using the second frequency position.

5 Claims, 10 Drawing Sheets



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3GPP TS 37.340 V16.8.0 (Dec. 2021) 3rd Generation Partnership Project; Technical Specification Group Radio Access Network; Evolved Universal Terrestrial Radio Access (E-UTRA) and NR; Multi-connectivity; Stage 2 (Release 16).
3GPP TS 38.211 V17.0.0 (Dec. 2021) 3rd Generation Partnership Project; Technical Specification Group Radio Access Network; NR; Physical channels and modulation (Release 17).
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3GPP TS 38.213 V17.0.0 (Dec. 2021) 3rd Generation Partnership Project; Technical Specification Group Radio Access Network; NR; Physical layer procedures for control (Release 17).
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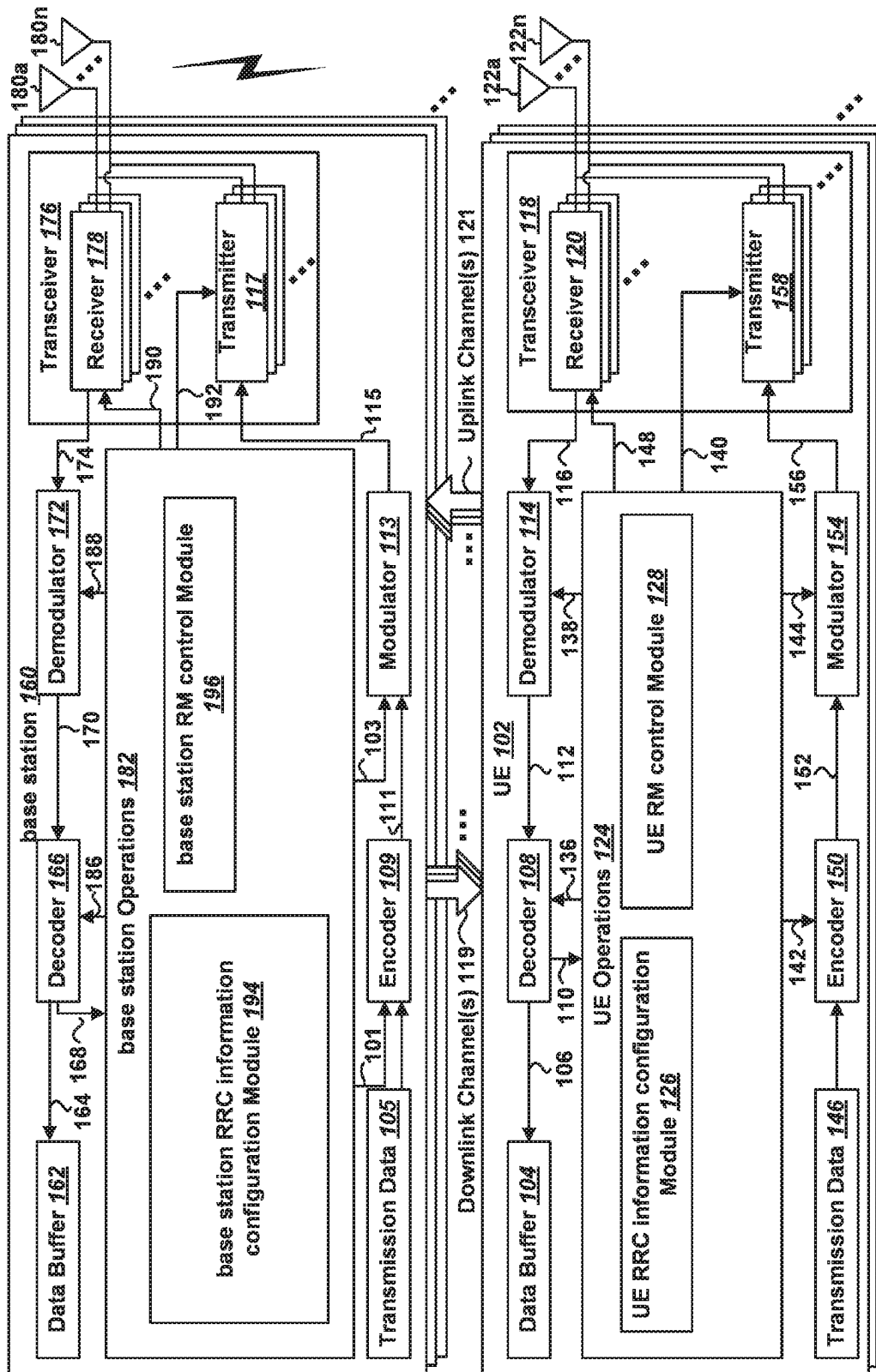


FIG. 1

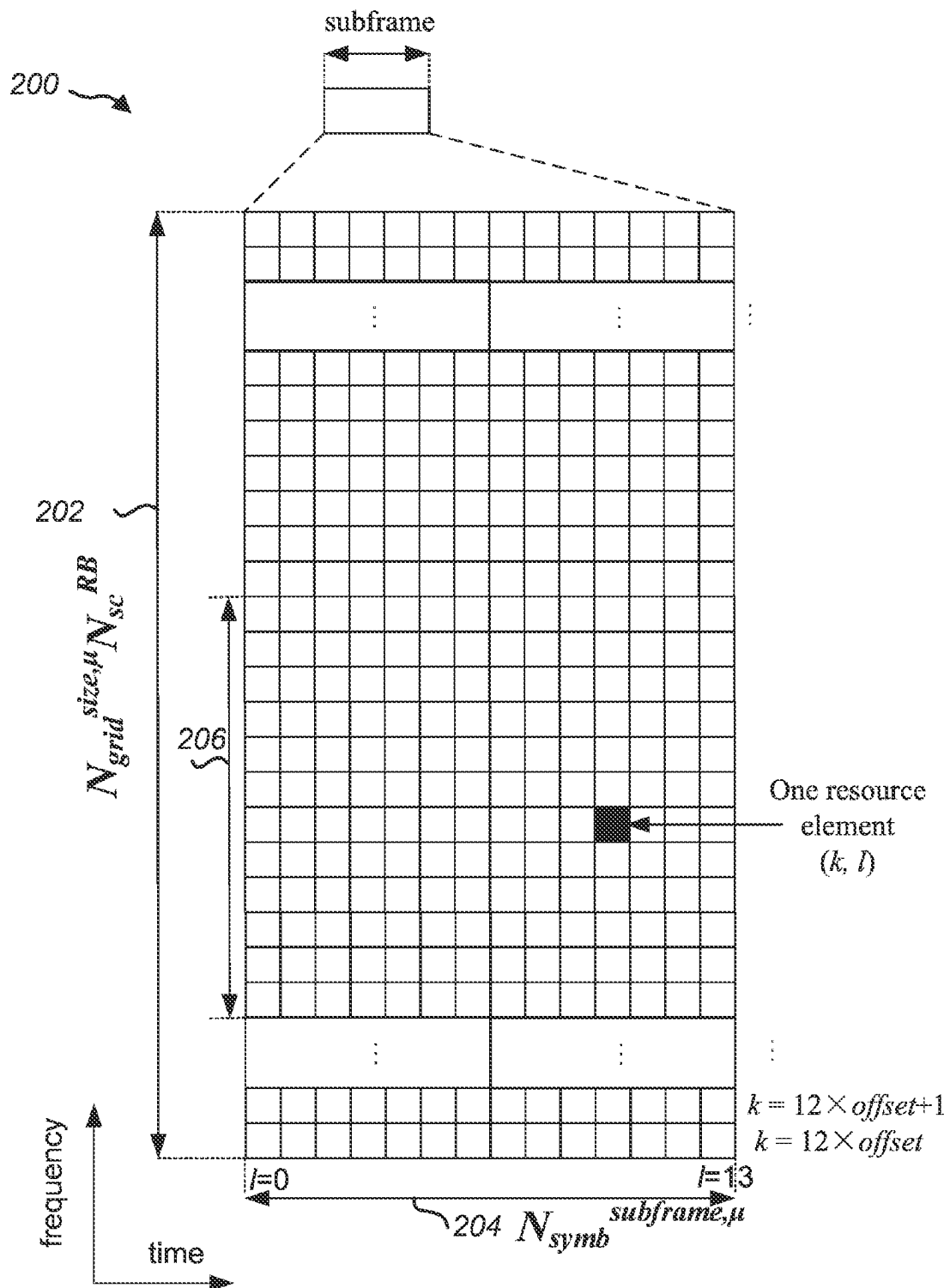


FIG. 2

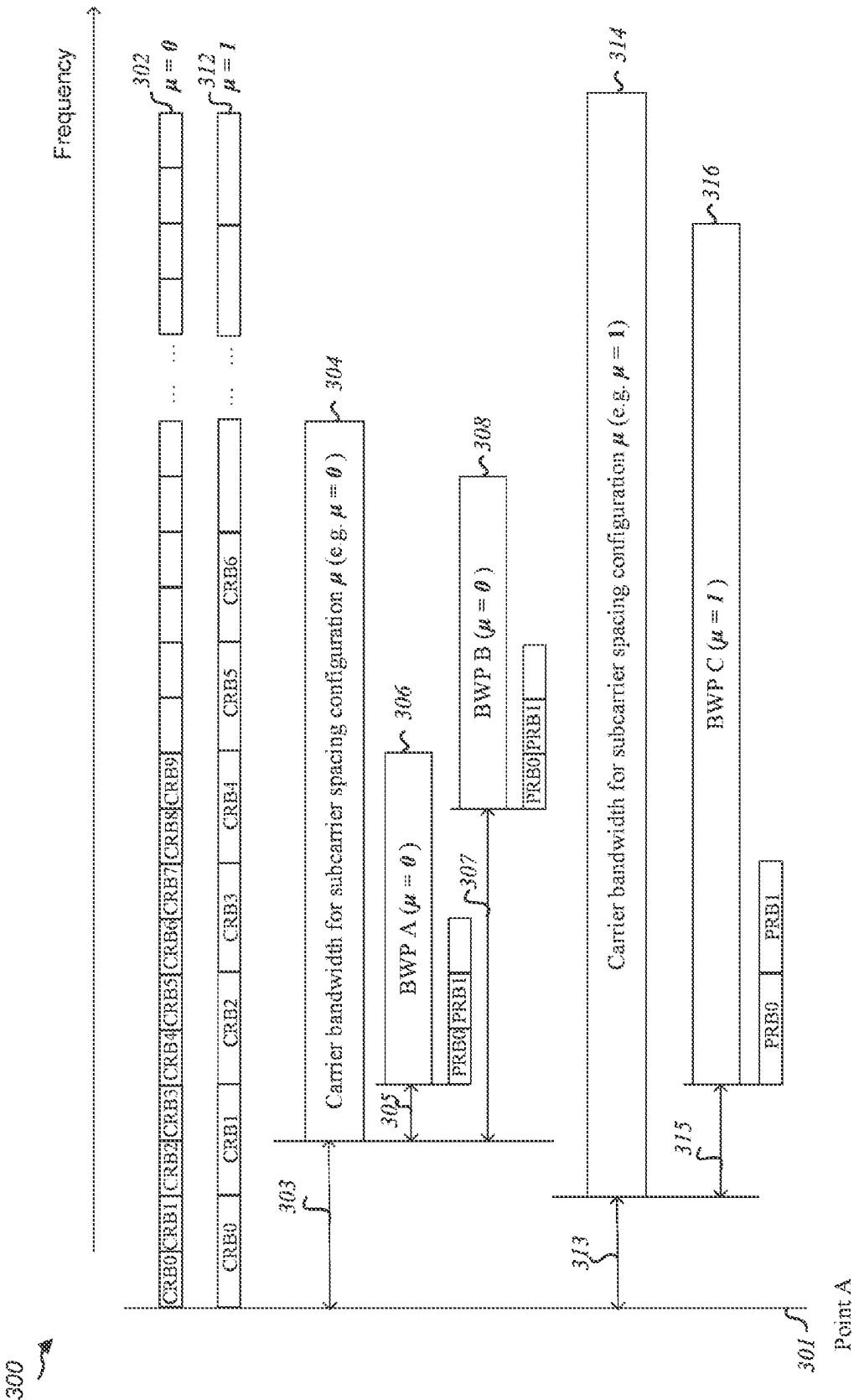


FIG. 3

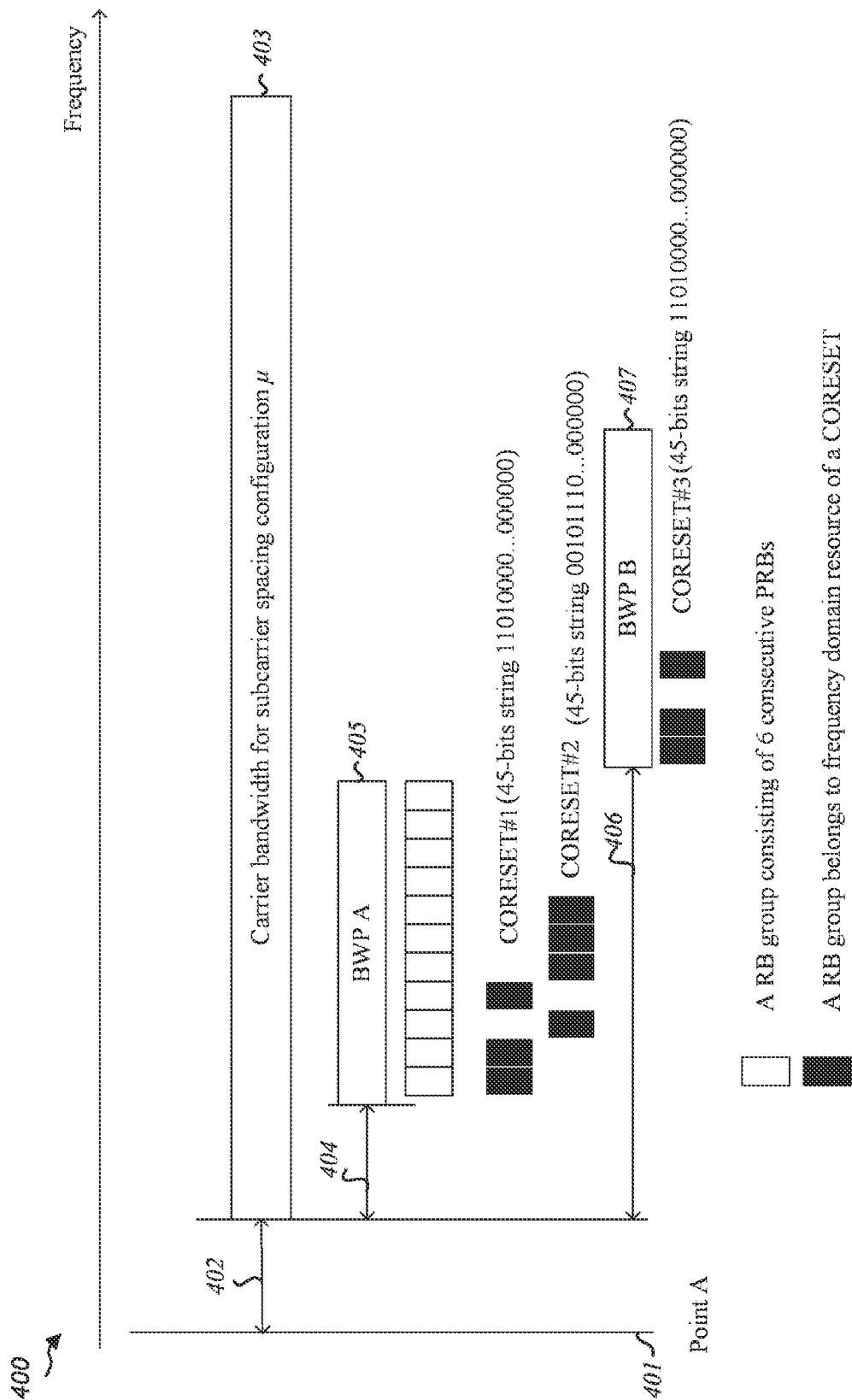
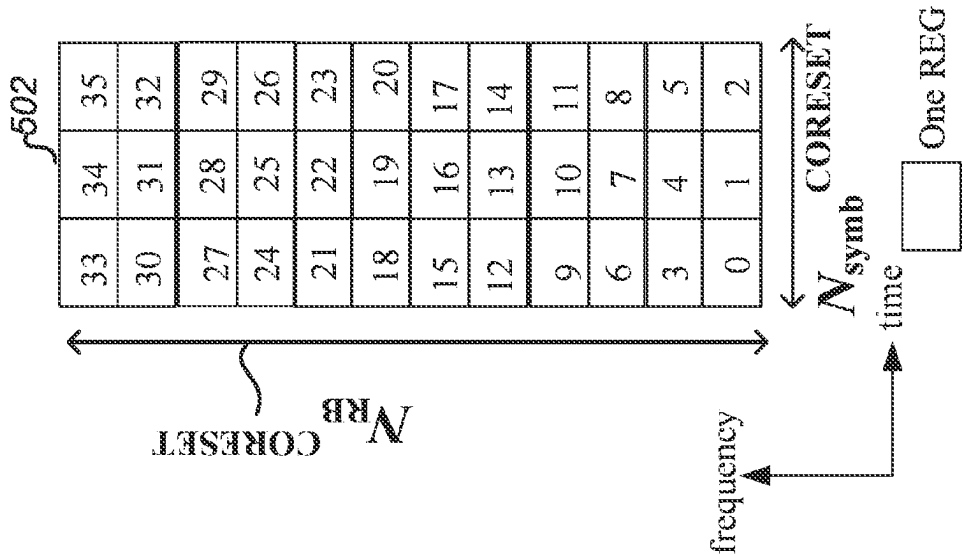


FIG. 4

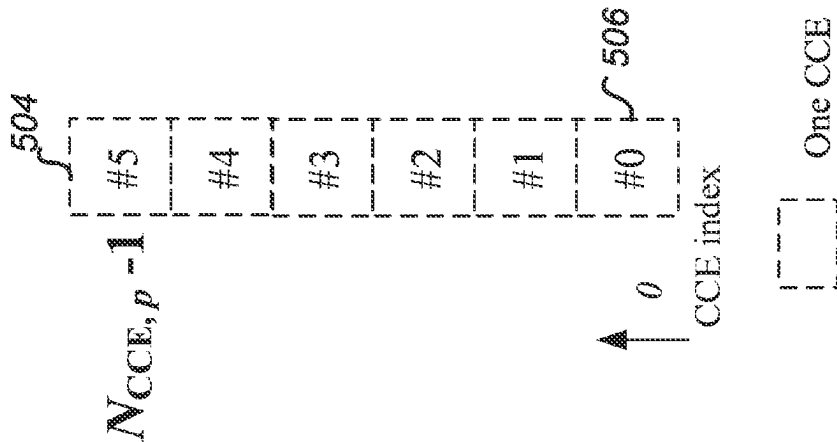
500-1

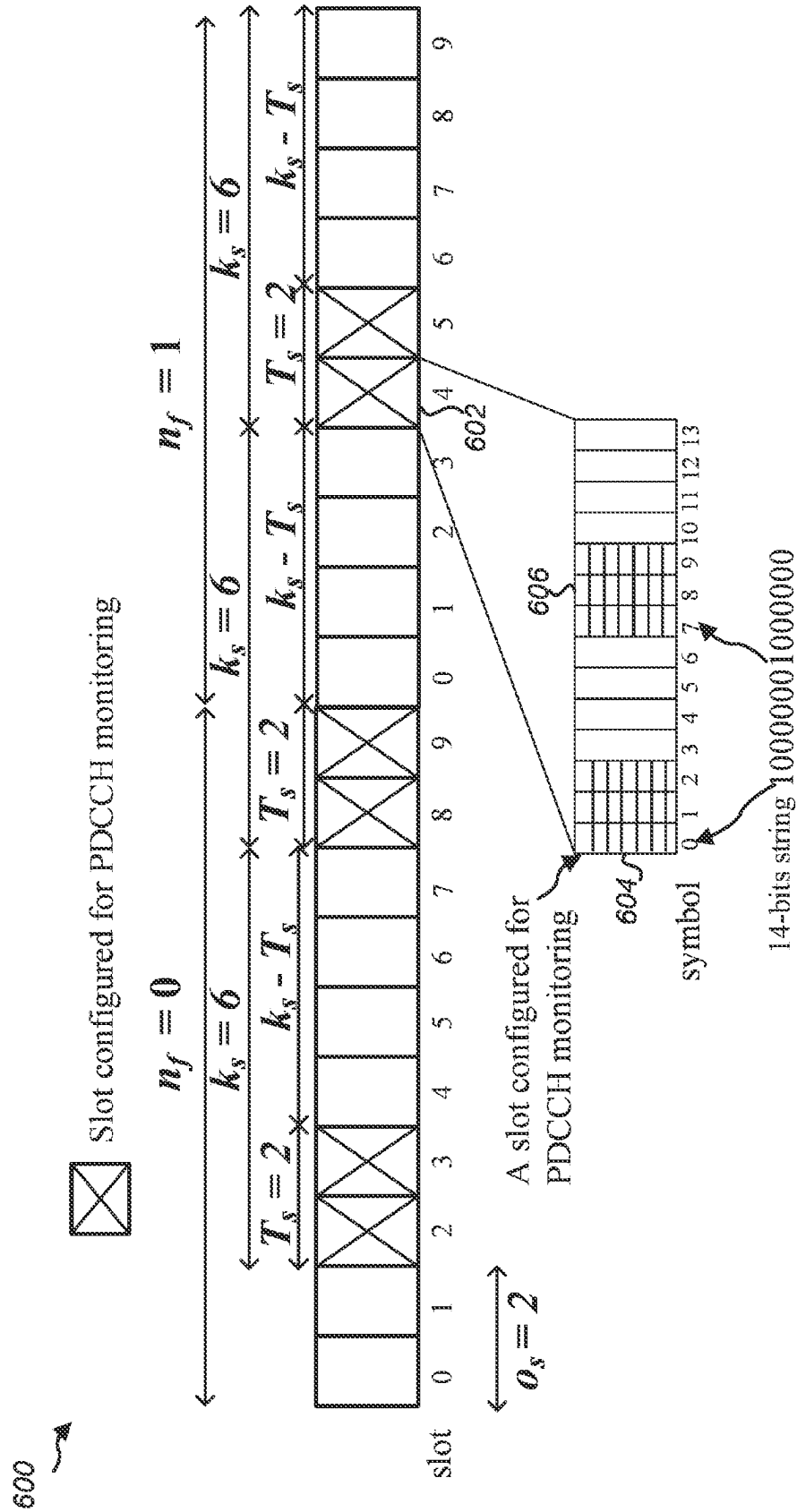
FIG. 5A



500-2

FIG. 5B





60

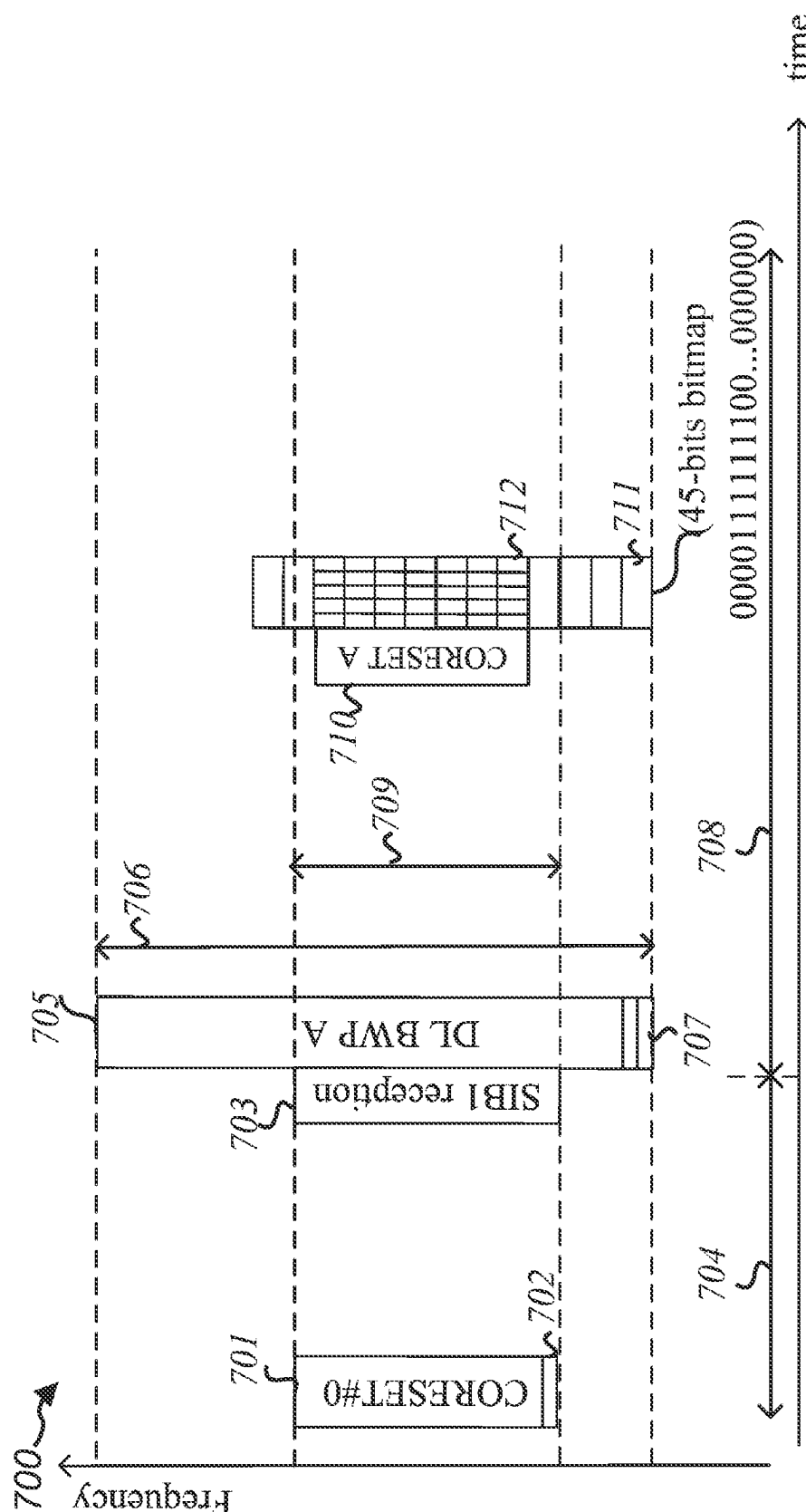
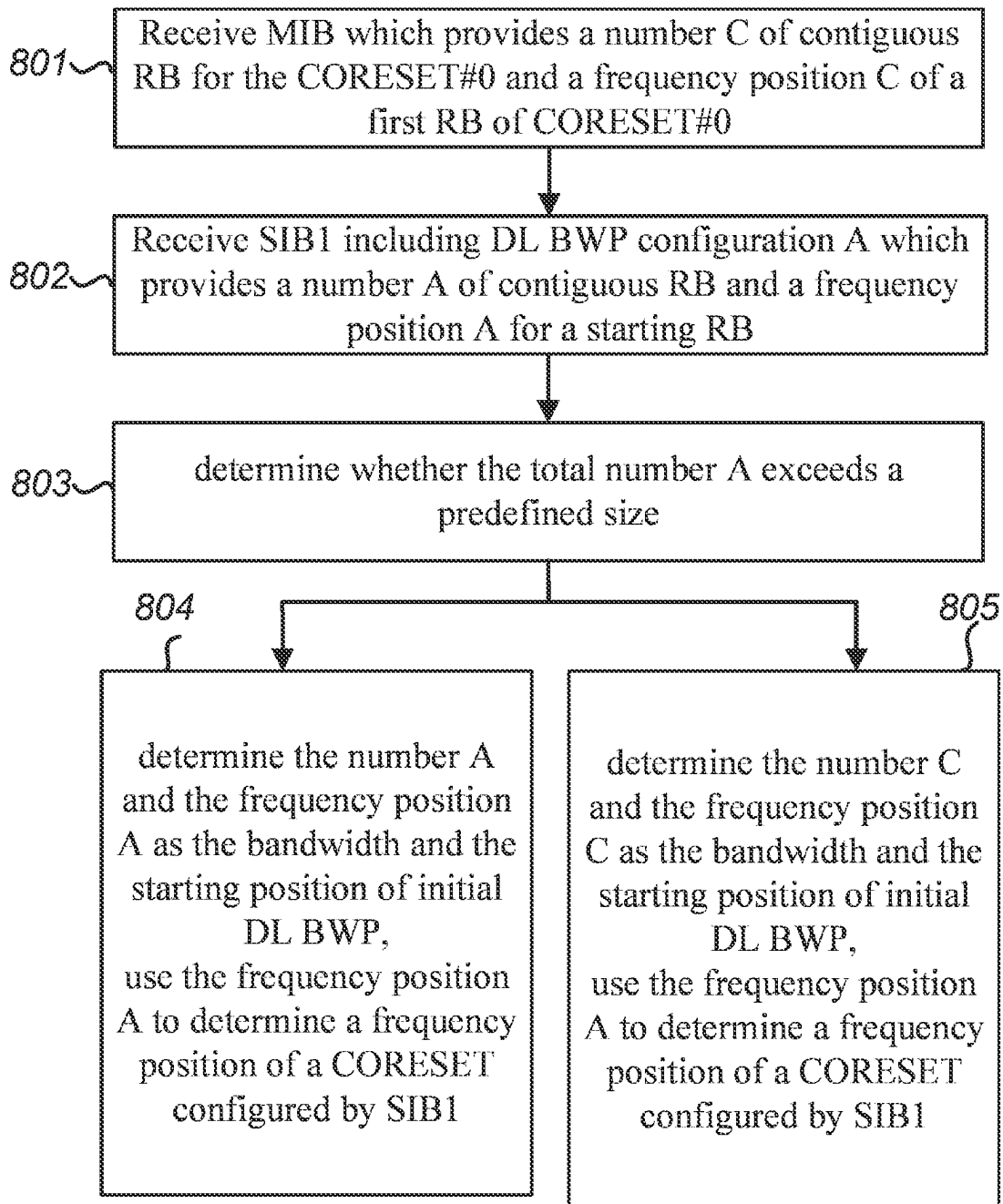


FIG. 7

800

FIG. 8

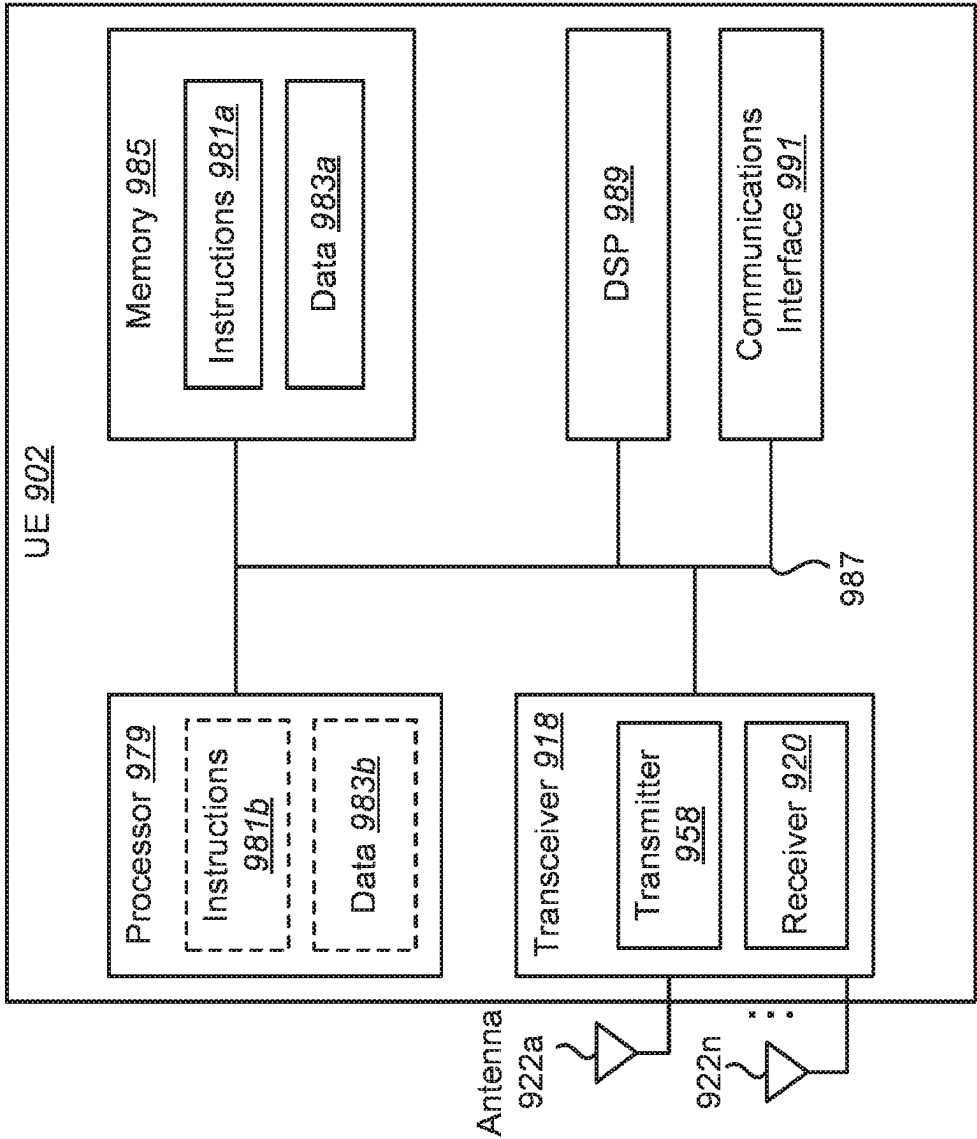


FIG. 9

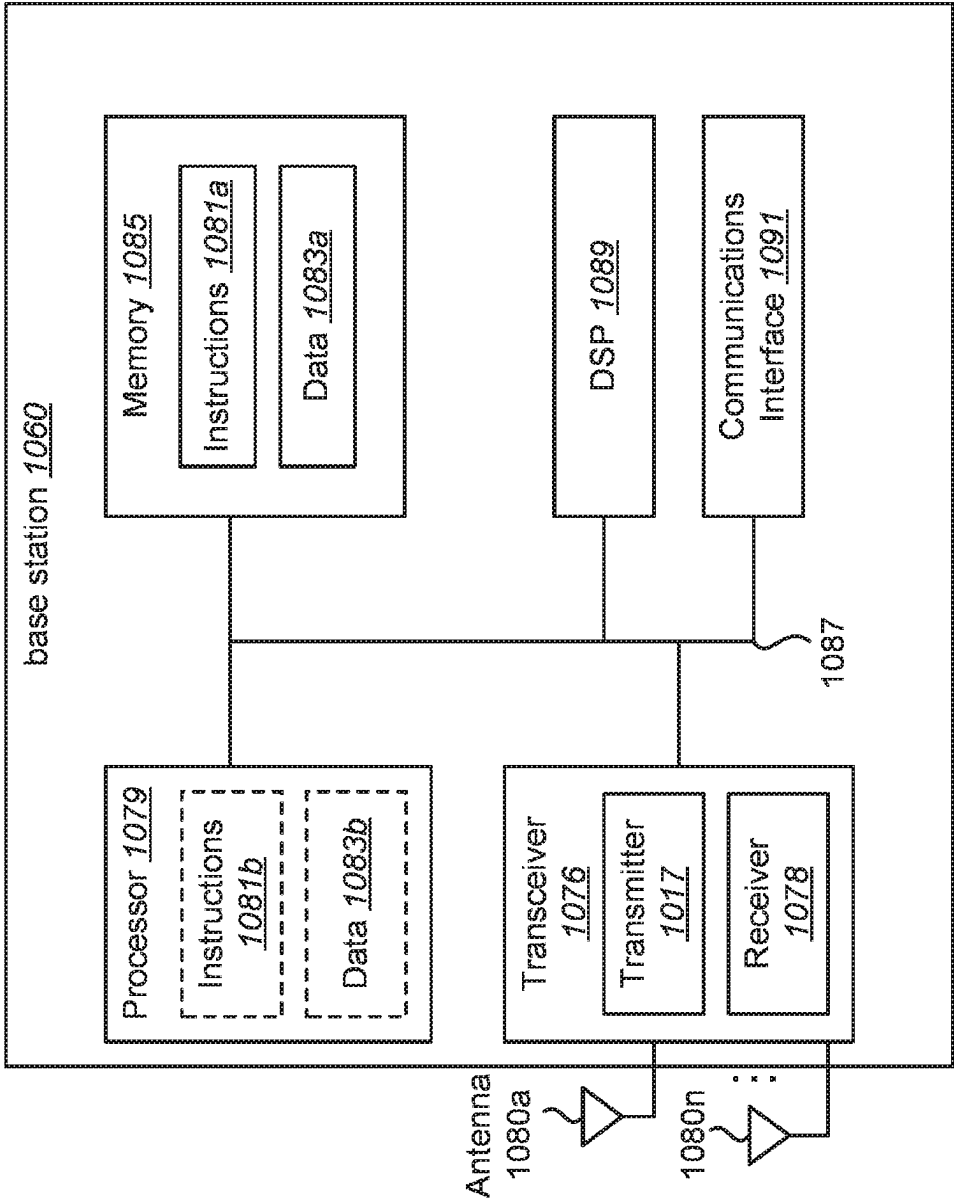


FIG. 10

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USER EQUIPMENT, BASE STATION, AND COMMUNICATION METHOD

TECHNICAL FIELD

The present disclosure relates to a user equipment, a base station, and a communication method.

BACKGROUND

At present, as a radio access system and a radio network technology aimed for the fifth generation cellular system, technical investigation and standard development are being conducted, as extended standards of Long Term Evolution (LTE), on LTE-Advanced Pro (LTE-A Pro) and New Radio technology (NR) in The Third Generation Partnership Project (3GPP).

In the fifth generation cellular system, three services of enhanced Mobile BroadBand (eMBB) to achieve high-speed and large-volume transmission, Ultra-Reliable and Low Latency Communication (URLLC) to achieve low-latency and high-reliability communication, and massive Machine Type Communication (mMTC) to allow connection of a large number of machine type devices such as Internet of Things (IoT) have been demanded as assumed scenarios.

For example, wireless communication devices may communicate with one or more devices for multiple service types. For some device types, a lower complexity would be required such as to reduce the Rx/Tx antennas and/or the RF/Baseband bandwidth to reduce the UE complexity and the UE cost. However, given the reduced antennas and/or the bandwidth, the flexibility and/or the efficiency of the whole system would be limited. As illustrated by this discussion, systems and methods according to the present invention, supporting how to determine initial downlink BWP and common CORESET from the initial DL BWP configurations, may improve the communication flexibility and efficiency and could be beneficial.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram illustrating one configuration of one or more base stations and one or more user equipments (UEs) in which systems and methods for how to determine initial DL BWP may be implemented;

FIG. 2 is a diagram illustrating one example 200 of a resource grid;

FIG. 3 is a diagram illustrating one example 300 of common resource block grid, carrier configuration and BWP configuration by a UE 102 and a base station 160;

FIG. 4 is a diagram illustrating one 400 example of CORESET configuration in a BWP by a UE 102 and a base station 160;

FIGS. 5A and 5B are diagrams illustrating one example 500-1 of REG numbering and one example 500-2 of CCE resource numbering for a CORESET;

FIG. 6 is a diagram illustrating one example 600 how to determine PDCCH monitoring occasions for PDCCH candidates based on corresponding search space set configuration and CORESET configuration;

FIG. 7 is a diagram illustrating one 700 example of initial DL BWP determination according to initial DL BWP configuration A and initial DL BWP configuration B by a UE 102 and a base station 160;

FIG. 8 is a flow diagram illustrating one implementation of a method 800 for initial DL BWP determination by a UE 102;

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FIG. 9 illustrates various components that may be utilized in a UE;

FIG. 10 illustrates various components that may be utilized in a base station;

DESCRIPTION OF EMBODIMENTS

A communication method by a user equipment (UE) is described. The method includes receiving, from a base station, from a base station, MIB providing a first quantity of contiguous resource blocks (RBs) and a first frequency position of a first starting RB for a first CORESET; receiving system information including a second configuration of an initial downlink (DL) BWP, the second configuration providing a second quantity of contiguous RBs, a second frequency position of a second starting RB, and a bitmap indicating frequency domain resources for a second CORESET, each bit of the bitmap corresponds a group of 6 contiguous RBs, with grouping starting from a first RB group in ascending order of RB index; and determining whether the second quantity exceeds a predefined size, wherein in a case the second quantity exceeds the predefined size, determining the first frequency position and the first quantity as a starting position and a bandwidth for an initial DL BWP, and determining a first RB index of the first RB group by using the second frequency position, in a case the second quantity does not exceed the predefined size, determining the second frequency position and the second quantity as the starting position and the bandwidth, and determining the first RB index of the first RB group by using the second frequency position.

A communication method by a base station is described. The method includes generating, to a user equipment (UE), a MIB to provide a first quantity of contiguous resource blocks (RBs) and a first frequency position of a first starting RB for a first CORESET; generating system information to include a second configuration of an initial downlink (DL) BWP, the second configuration providing a second quantity of contiguous RBs, a second frequency position of a second starting RB, and a bitmap indicating frequency domain resources for a second CORESET, each bit of the bitmap corresponds a group of 6 contiguous RBs, with grouping starting from a first RB group in ascending order of RB index; determining whether the second quantity exceeds a predefined size, wherein, in a case the second quantity exceeds the predefined size, determining the first frequency position and the first quantity as a starting position and a bandwidth for an initial DL BWP of the UE, and determining a first RB index of the first RB group by using the second frequency position, in a case the second quantity does not exceed the predefined size, determining the second frequency position and the second quantity as the starting position and the bandwidth, and determining the first RB index of the first RB group by using the second frequency position; and transmitting the MIB and the system information.

A user equipment (UE) is described. The UE includes reception circuitry configured to receive, from a base station, MIB providing a first quantity of contiguous resource blocks (RBs) and a first frequency position of a first starting RB for a first CORESET, and to receive system information including a second configuration of an initial downlink (DL) BWP, the second configuration providing a second quantity of contiguous RBs, a second frequency position of a second starting RB, and a bitmap indicating frequency domain resources for a second CORESET, each bit of the bitmap corresponds a group of 6 contiguous RBs, with grouping

starting from a first RB group in ascending order of RB index; and control circuitry configured to determine whether the second quantity exceeds a predefined size, wherein the control circuitry is further configured to, in a case the second quantity exceeds the predefined size, determine the first frequency position and the first quantity as a starting position and a bandwidth for an initial DL BWP, and determine a first RB index of the first RB group by using the second frequency position, in a case the second quantity does not exceed the predefined size, determine the second frequency position and the second quantity as the starting position and the bandwidth, and determine the first RB index of the first RB group by using the second frequency position.

A base station is described. The base station includes transmission circuitry configured to generate, to a user equipment (UE), a MIB to provide a first quantity of contiguous resource blocks (RBs) and a first frequency position of a first starting RB for a first CORESET, to generate system information to include a second configuration of an initial downlink (DL) BWP, the second configuration providing a second quantity of contiguous RBs, a second frequency position of a second starting RB, and a bitmap indicating frequency domain resources for a second CORESET, each bit of the bitmap corresponds a group of 6 contiguous RBs, with grouping starting from a first RB group in ascending order of RB index, to determine whether the second quantity exceeds a predefined size, wherein the control circuitry is further configured to, in a case the second quantity exceeds the predefined size, determine the first frequency position and the first quantity as a starting position and a bandwidth for an initial DL BWP of the UE, and determine a first RB index of the first RB group by using the second frequency position, in a case the second quantity does not exceed the predefined size, determine the second frequency position and the second quantity as the starting position and the bandwidth, and determine the first RB index of the first RB group by using the second frequency position; and transmission circuitry configured to transmit the MIB and the system information.

3GPP Long Term Evolution (LTE) is the name given to a project to improve the Universal Mobile Telecommunications System (UMTS) mobile phone or device standard to cope with future requirements. In one aspect, UMTS has been modified to provide support and specification for the Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Universal Terrestrial Radio Access Network (E-UTRAN). 3GPP NR (New Radio) is the name given to a project to improve the LTE mobile phone or device standard to cope with future requirements. In one aspect, LTE has been modified to provide support and specification (TS 38.331, 38.321, 38.300, 37.340, 38.211, 38.212, 38.213, 38.214, etc.) for the New Radio Access (NR) and Next generation—Radio Access Network (NG-RAN).

At least some aspects of the systems and methods disclosed herein may be described in relation to the 3GPP LTE, LTE-Advanced (LTE-A), LTE-Advanced Pro, New Radio Access (NR), and other 3G/4G/5G standards (e.g., 3GPP Releases 8, 9, 10, 11, 12, 13, 14, 15, and/or 16, and/or Narrow Band-Internet of Things (NB-IoT)). However, the scope of the present disclosure should not be limited in this regard. At least some aspects of the systems and methods disclosed herein may be utilized in other types of wireless communication systems.

A wireless communication device may be an electronic device used to communicate voice and/or data to a base station, which in turn may communicate with a network of devices (e.g., public switched telephone network (PSTN),

the Internet, etc.). In describing systems and methods herein, a wireless communication device may alternatively be referred to as a mobile station, a UE (User Equipment), an access terminal, a subscriber station, a mobile terminal, a remote station, a user terminal, a terminal, a subscriber unit, a mobile device, a relay node, etc. Examples of wireless communication devices include cellular phones, smart phones, personal digital assistants (PDAs), laptop computers, netbooks, e-readers, wireless modems, industrial wireless sensors, video surveillance, wearables, etc. In 3GPP specifications, a wireless communication device is typically referred to as a UE. However, as the scope of the present disclosure should not be limited to the 3GPP standards, the terms “UE” and “wireless communication device” may be used interchangeably herein to mean the more general term “wireless communication device.”

In 3GPP specifications, a base station is typically referred to as a gNB, a Node B, an eNB, a home enhanced or evolved Node B (HeNB) or some other similar terminology. As the scope of the disclosure should not be limited to 3GPP standards, the terms “base station,” “gNB,” “Node B,” “eNB,” and “HeNB” may be used interchangeably herein to mean the more general term “base station.” Furthermore, one example of a “base station” is an access point. An access point may be an electronic device that provides access to a network (e.g., Local Area Network (LAN), the Internet, etc.) for wireless communication devices. The term “communication device” may be used to denote both a wireless communication device and/or a base station.

It should be noted that as used herein, a “cell” may be any communication channel that is specified by standardization or regulatory bodies to be used for International Mobile Telecommunications-Advanced (IMT-Advanced), IMT-2020 (5G) and all of it or a subset of it may be adopted by 3GPP as licensed bands (e.g., frequency bands) to be used for communication between a base station and a UE. It should also be noted that in NR, NG-RAN, E-UTRA and E-UTRAN overall description, as used herein, a “cell” may be defined as “combination of downlink and optionally uplink resources.” The linking between the carrier frequency of the downlink resources and the carrier frequency of the uplink resources may be indicated in the system information transmitted on the downlink resources.

“Configured cells” are those cells of which the UE is aware and is allowed by a base station to transmit or receive information. “Configured cell(s)” may be serving cell(s). The UE may receive system information and perform the required measurements on configured cells. “Configured cell(s)” for a radio connection may consist of a primary cell and/or no, one, or more secondary cell(s). “Activated cells” are those configured cells on which the UE is transmitting and receiving. That is, activated cells are those cells for which the UE monitors the physical downlink control channel (PDCCH) and in the case of a downlink transmission, those cells for which the UE decodes a physical downlink shared channel (PDSCH). “Deactivated cells” are those configured cells that the UE is not monitoring the transmission PDCCH. It should be noted that a “cell” may be described in terms of differing dimensions. For example, a “cell” may have temporal, spatial (e.g., geographical) and frequency characteristics.

The base stations may be connected by the NG interface to the 5G-core network (5G-CN). 5G-CN may be called as to NextGen core (NGC), or 5G core (5GC). The base stations may also be connected by the S1 interface to the evolved packet core (EPC). For instance, the base stations may be connected to a NextGen (NG) mobility management

function by the NG-2 interface and to the NG core User Plane (UP) functions by the NG-3 interface. The NG interface supports a many-to-many relation between NG mobility management functions, NG core UP functions and the base stations. The NG-2 interface is the NG interface for the control plane and the NG-3 interface is the NG interface for the user plane. For instance, for EPC connection, the base stations may be connected to a mobility management entity (MME) by the S1-MME interface and to the serving gateway (S-GW) by the S1-U interface. The S1 interface supports a many-to-many relation between MMES, serving gateways and the base stations. The S1-MME interface is the S1 interface for the control plane and the S1-U interface is the S1 interface for the user plane. The Uu interface is a radio interface between the UE and the base station for the radio protocol.

The radio protocol architecture may include the user plane and the control plane. The user plane protocol stack may include packet data convergence protocol (PDCP), radio link control (RLC), medium access control (MAC) and physical (PHY) layers. A DRB (Data Radio Bearer) is a radio bearer that carries user data (as opposed to control plane signaling). For example, a DRB may be mapped to the user plane protocol stack. The PDCP, RLC, MAC and PHY sublayers (terminated at the base station 460a on the network) may perform functions (e.g., header compression, ciphering, scheduling, ARQ and HARQ) for the user plane. PDCP entities are located in the PDCP sublayer. RLC entities may be located in the RLC sublayer. MAC entities may be located in the MAC sublayer. The PHY entities may be located in the PHY sublayer.

The control plane may include a control plane protocol stack. The PDCP sublayer (terminated in base station on the network side) may perform functions (e.g., ciphering and integrity protection) for the control plane. The RLC and MAC sublayers (terminated in base station on the network side) may perform the same functions as for the user plane. The Radio Resource Control (RRC) (terminated in base station on the network side) may perform the following functions. The RRC may perform broadcast functions, paging, RRC connection management, radio bearer (RB) control, mobility functions, UE measurement reporting and control. The Non-Access Stratum (NAS) control protocol (terminated in MME on the network side) may perform, among other things, evolved packet system (EPS) bearer management, authentication, evolved packet system connection management (ECM)-IDLE mobility handling, paging origination in ECM-IDLE and security control.

Signaling Radio Bearers (SRBs) are Radio Bearers (RB) that may be used only for the transmission of RRC and NAS messages. Three SRBs may be defined. SRB0 may be used for RRC messages using the common control channel (CCCH) logical channel. SRB1 may be used for RRC messages (which may include a piggybacked NAS message) as well as for NAS messages prior to the establishment of SRB2, all using the dedicated control channel (DCCH) logical channel. SRB2 may be used for RRC messages which include logged measurement information as well as for NAS messages, all using the DCCH logical channel. SRB2 has a lower-priority than SRB1 and may be configured by a network (e.g., base station) after security activation. A broadcast control channel (BCCH) logical channel may be used for broadcasting system information. Some of BCCH logical channel may convey system information which may be sent from the network to the UE via BCH (Broadcast Channel) transport channel. BCH may be sent on a physical broadcast channel (PBCH). Some of BCCH

logical channel may convey system information which may be sent from the network to the UE via DL-SCH (Downlink Shared Channel) transport channel. Paging may be provided by using paging control channel (PCCH) logical channel.

For example, the DL-DCCH logical channel may be used (but not limited to) for a RRC reconfiguration message, a RRC reestablishment message, a RRC release, a UE Capability Enquiry message, a DL Information Transfer message or a Security Mode Command message. UL-DCCH logical channel may be used (but not limited to) for a measurement report message, a RRC Reconfiguration Complete message, a RRC Reestablishment Complete message, a RRC Setup Complete message, a Security Mode Complete message, a Security Mode Failure message, a UE Capability Information message, a UL Handover Preparation Transfer message, a UL Information Transfer message, a Counter Check Response message, a UE Information Response message, a Proximity Indication message, a RN (Relay Node) Reconfiguration Complete message, an MBMS Counting Response message, an inter Frequency RSTD Measurement Indication message, a UE Assistance Information message, an In-device Coexistence Indication message, an MBMS Interest Indication message, an SCG Failure Information message. DL-CCCH logical channel may be used (but not limited to) for a RRC Connection Reestablishment message, a RRC Reestablishment Reject message, a RRC Reject message, or a RRC Setup message. UL-CCCH logical channel may be used (but not limited to) for a RRC Reestablishment Request message, or a RRC Setup Request message.

System information may be divided into the MasterInformationBlock (MIB) and a number of SystemInformationBlocks (SIBs).

The UE may receive one or more RRC messages from the base station to obtain RRC configurations or parameters. The RRC layer of the UE may configure RRC layer and/or lower layers (e.g., PHY layer, MAC layer, RLC layer, PDCP layer) of the UE according to the RRC configurations or parameters which may be configured by the RRC messages, broadcasted system information, and so on. The base station may transmit one or more RRC messages to the UE to cause the UE to configure RRC layer and/or lower layers of the UE according to the RRC configurations or parameters which may be configured by the RRC messages, broadcasted system information, and so on.

When carrier aggregation is configured, the UE may have one RRC connection with the network. One radio interface may provide carrier aggregation. During RRC establishment, re-establishment and handover, one serving cell may provide Non-Access Stratum (NAS) mobility information (e.g., a tracking area identity (TAI)). During RRC re-establishment and handover, one serving cell may provide a security input. This cell may be referred to as the primary cell (PCell). In the downlink, the component carrier corresponding to the PCell may be the downlink primary component carrier (DL PCC), while in the uplink it may be the uplink primary component carrier (UL PCC). In the present disclosure, the terms "component carrier" and "carrier" can be interchanged with each other.

Depending on UE capabilities, one or more SCells may be configured to form together with the PCell a set of serving cells. In the downlink, the component carrier corresponding to an SCell may be a downlink secondary component carrier (DL SCC), while in the uplink it may be an uplink secondary component carrier (UL SCC).

The configured set of serving cells for the UE, therefore, may consist of one PCell and one or more SCells. For each

SCell, the usage of uplink resources by the UE (in addition to the downlink resources) may be configurable. The number of DL SCCs configured may be larger than or equal to the number of UL SCCs and no SCell may be configured for usage of uplink resources only.

From a UE viewpoint, each uplink resource may belong to one serving cell. The number of serving cells that may be configured depends on the aggregation capability of the UE. The PCell may only be changed using a handover procedure (e.g., with a security key change and a random access procedure). A PCell may be used for transmission of the PUCCH. A primary secondary cell (PSCell) may also be used for transmission of the PUCCH. The PSCell may be referred to as a primary SCG cell or SpCell of a secondary cell group. The PCell or PSCell may not be de-activated. Re-establishment may be triggered when the PCell experiences radio link failure (RLF), not when the SCells experience RLF. Furthermore, NAS information may be taken from the PCell.

The reconfiguration, addition and removal of SCells may be performed by RRC. At handover or reconfiguration with sync, Radio Resource Control (RRC) layer may also add, remove or reconfigure SCells for usage with a target PCell. When adding a new SCell, dedicated RRC signaling may be used for sending all required system information of the SCell (e.g., while in connected mode, UEs need not acquire broadcasted system information directly from the SCells).

The systems and methods described herein may enhance the efficient use of radio resources in Carrier aggregation (CA) operation. Carrier aggregation refers to the concurrent utilization of more than one component carrier (CC). In carrier aggregation, more than one cell may be aggregated to a UE. In one example, carrier aggregation may be used to increase the effective bandwidth available to a UE. In traditional carrier aggregation, a single base station is assumed to provide multiple serving cells for a UE. Even in scenarios where two or more cells may be aggregated (e.g., a macro cell aggregated with remote radio head (RRH) cells) the cells may be controlled (e.g., scheduled) by a single base station. However, in a small cell deployment scenario, each node (e.g., base station, RRH, etc.) may have its own independent scheduler. To maximize the efficiency of radio resources utilization of both nodes, a UE may connect to two or more nodes that have different schedulers. The systems and methods described herein may enhance the efficient use of radio resources in dual connectivity operation. A UE may be configured multiple groups of serving cells, where each group may have carrier aggregation operation (e.g., if the group includes more than one serving cell).

In Dual Connectivity (DC) the UE may be required to be capable of UL-CA with simultaneous PUCCH/PUSCH and PUCCH/PUSCH transmissions across cell-groups (CGs). In a small cell deployment scenario, each node (e.g., eNB, RRH, etc.) may have its own independent scheduler. To maximize the efficiency of radio resources utilization of both nodes, a UE may connect to two or more nodes that have different schedulers. A UE may be configured multiple groups of serving cells, where each group may have carrier aggregation operation (e.g., if the group includes more than one serving cell). A UE in RRC_CONNECTED may be configured with Dual Connectivity or MR-DC, when configured with a Master and a Secondary Cell Group. A Cell Group (CG) may be a subset of the serving cells of a UE, configured with Dual Connectivity (DC) or MR-DC, i.e. a Master Cell Group (MCG) or a Secondary Cell Group (SCG). The Master Cell Group may be a group of serving cells of a UE comprising of the PCell and zero or more

secondary cells. The Secondary Cell Group (SCG) may be a group of secondary cells of a UE, configured with DC or MR-DC, comprising of the PSCell and zero or more other secondary cells. A Primary Secondary Cell (PSCell) may be the SCG cell in which the UE is instructed to perform random access when performing the SCG change procedure. "PSCell" may be also called as a Primary SCG Cell. In Dual Connectivity or MR-DC, two MAC entities may be configured in the UE: one for the MCG and one for the SCG. Each MAC entity may be configured by RRC with a serving cell supporting PUCCH transmission and contention based Random Access. In a MAC layer, the term Special Cell (SpCell) may refer to such cell, whereas the term SCell may refer to other serving cells. The term SpCell either may refer to the PCell of the MCG or the PSCell of the SCG depending on if the MAC entity is associated to the MCG or the SCG, respectively. A Timing Advance Group (TAG) containing the SpCell of a MAC entity may be referred to as primary TAG (pTAG), whereas the term secondary TAG (sTAG) refers to other TAGs.

DC may be further enhanced to support Multi-RAT Dual Connectivity (MR-DC). MR-DC may be a generalization of the Intra-E-UTRA Dual Connectivity (DC) described in 36.300, where a multiple Rx/Tx UE may be configured to utilize resources provided by two different nodes connected via non-ideal backhaul, one providing E-UTRA access and the other one providing NR access. One node acts as a Master Node (MN) and the other as a Secondary Node (SN). The MN and SN are connected via a network interface and at least the MN is connected to the core network. In DC, a PSCell may be a primary secondary cell. In EN-DC, a PSCell may be a primary SCG cell or SpCell of a secondary cell group.

E-UTRAN may support MR-DC via E-UTRA-NR Dual Connectivity (EN-DC), in which a UE is connected to one eNB that acts as a MN and one en-gNB that acts as a SN. The en-gNB is a node providing NR user plane and control plane protocol terminations towards the UE, and acting as Secondary Node in EN-DC. The eNB is connected to the EPC via the S1 interface and to the en-gNB via the X2 interface. The en-gNB might also be connected to the EPC via the S1-U interface and other en-gNBs via the X2-U interface.

A timer is running once it is started, until it is stopped or until it expires; otherwise it is not running. A timer can be started if it is not running or restarted if it is running. A Timer may be always started or restarted from its initial value.

For NR, a technology of aggregating NR carriers may be studied. Both lower layer aggregation like Carrier Aggregation (CA) for LTE and upper layer aggregation like DC are investigated. From layer 2/3 point of view, aggregation of carriers with different numerologies may be supported in NR.

The main services and functions of the RRC sublayer may include the following:

- Broadcast of System Information related to Access Stratum (AS) and Non Access Stratum (NAS);
- Paging initiated by CN or RAN;
- Establishment, maintenance and release of an RRC connection between the UE and NR RAN including:
 - Addition, modification and release of carrier aggregation;
 - Addition, modification and release of Dual Connectivity in NR or between LTE and NR;
- Security functions including key management;
- Establishment, configuration, maintenance and release of signaling radio bearers and data radio bearers;

Mobility functions including:

Handover;

UE cell selection and reselection and control of cell selection and reselection;

Context transfer at handover.

QoS management functions;

UE measurement reporting and control of the reporting;

NAS message transfer to/from NAS from/to UE.

Each MAC entity of a UE may be configured by RRC with a Discontinuous Reception (DRX) functionality that controls the UE's PDCCH monitoring activity for the MAC entity's C-RNTI (Radio Network Temporary Identifier), CS-RNTI, INT-RNTI, SFI-RNTI, SP-CSI-RNTI, TPC-PUCCH-RNTI, TPC-PUSCH-RNTI, and TPC-SRS-RNTI. For scheduling at cell level, the following identities are used:

C (Cell)-RNTI: unique UE identification used as an identifier of the RRC Connection and for scheduling;

CS (Configured Scheduling)-RNTI: unique UE identification used for Semi-Persistent Scheduling in the downlink;

INT-RNTI: identification of pre-emption in the downlink;

P-RNTI: identification of Paging and System Information change notification in the downlink;

SI-RNTI: identification of Broadcast and System Information in the downlink;

SP-CSI-RNTI: unique UE identification used for semi-persistent CSI reporting on PUSCH;

CI-RNTI: Cancellation Indication RNTI for Uplink.

For power and slot format control, the following identities are used:

SFI-RNTI: identification of slot format;

TPC-PUCCH-RNTI: unique UE identification to control the power of PUCCH;

TPC-PUSCH-RNTI: unique UE identification to control the power of PUSCH;

TPC-SRS-RNTI: unique UE identification to control the power of SRS;

During the random access procedure, the following identities are also used:

RA-RNTI: identification of the Random Access Response in the downlink;

Temporary C-RNTI: UE identification temporarily used for scheduling during the random access procedure;

Random value for contention resolution: UE identification temporarily used for contention resolution purposes during the random access procedure.

For NR connected to 5GC, the following UE identities are used at NG-RAN level:

I-RNTI: used to identify the UE context in RRC_INACTIVE.

The size of various fields in the time domain is expressed in time units $T_c = 1/(\Delta f_{max} \times N_f)$ where $\Delta f_{max} = 480 \times 10^3$ Hz and $N_f = 4096$. The constant $\kappa = T_s/T_c = 64$ where $T_s = 1/(\Delta f_{ref} \times N_{f,ref})$, $\Delta f_{ref} = 15 \times 10^3$ Hz and $N_{f,ref} = 2048$.

Multiple OFDM numerologies are supported as given by Table 4.2-1 of [TS 38.211] where μ and the cyclic prefix for a bandwidth part are obtained from the higher-layer parameter subcarrierSpacing and cyclicPrefix, respectively.

The size of various fields in the time domain may be expressed as a number of time units $T_c = 1/(15000 \times 2048)$ seconds. Downlink and uplink transmissions are organized into frames with $T_f = (\Delta f_{max} N_f / 100) \cdot T_c = 10$ ms duration, each consisting of ten subframes of $T_s = (\Delta f_{max} N_f / 1000) \cdot T_c = 1$ ms duration. The number of consecutive OFDM symbols per subframe is $N_{symb}^{subframe, \mu} = N_{symb}^{slot} N_{slot}^{subframe, \mu}$. Each frame is divided into two equally-sized half-frames of five

subframes each with half-frame 0 consisting of subframes 0-4 and half-frame 1 consisting of subframes 5-9.

For subcarrier spacing (SCS) configuration μ , slots are numbered $n_s^\mu \in \{0, \dots, N_{slot}^{subframe, \mu} - 1\}$ in increasing order within a subframe and $n_{s,f}^\mu \in \{0, \dots, N_{slot}^{frame, \mu} - 1\}$ in increasing order within a frame. $N_{slot}^{subframe, \mu}$ is the number of slots per subframe for subcarrier spacing configuration μ . There are N_{symb}^{slot} consecutive OFDM symbols in a slot where N_{symb}^{slot} depends on the cyclic prefix as given by Tables 4.3.2-1 and 4.3.2-2 of [TS 38.211]. The start of slot n_s^μ in a subframe is aligned in time with the start of OFDM symbol $n_s^\mu N_{symb}^{slot}$ in the same subframe. Subcarrier spacing refers to a spacing (or frequency bandwidth) between two consecutive subcarrier in the frequency domain. For example, the subcarrier spacing can be set to 15 kHz (i.e. $\mu=0$), 30 kHz (i.e. $\mu=1$), 60 kHz (i.e. $\mu=2$), 120 kHz (i.e. $\mu=3$), or 240 kHz (i.e. $\mu=4$). A resource block is defined as a number of consecutive subcarriers (e.g. 12) in the frequency domain. For a carrier with different frequency, the applicable subcarrier may be different. For example, for a carrier in a frequency range 1, a subcarrier spacing only among a set of {15 kHz, 30 kHz, 60 kHz} is applicable. For a carrier in a frequency range 2, a subcarrier spacing only among a set of {60 kHz, 120 kHz, 240 kHz} is applicable. The base station may not configure an inapplicable subcarrier spacing for a carrier.

OFDM symbols in a slot can be classified as 'downlink', 'flexible', or 'uplink'. Signaling of slot formats is described in subclause 11.1 of [TS 38.213].

In a slot in a downlink frame, the UE may assume that downlink transmissions only occur in 'downlink' or 'flexible' symbols. In a slot in an uplink frame, the UE may only transmit in 'uplink' or 'flexible' symbols.

Various examples of the systems and methods disclosed herein are now described with reference to the Figures, where like reference numbers may indicate functionally similar elements. The systems and methods as generally described and illustrated in the Figures herein could be arranged and designed in a wide variety of different implementations. Thus, the following more detailed description of several implementations, as represented in the Figures, is not intended to limit scope, as claimed, but is merely representative of the systems and methods.

FIG. 1 is a block diagram illustrating one configuration of one or more base stations 160 (e.g., eNB, gNB) and one or more user equipments (UEs) 102 in which systems and methods for how to determine initial DL BWP may be implemented. The one or more UEs 102 may communicate with one or more base stations 160 using one or more antennas 122a-n. For example, a UE 102 transmits electromagnetic signals to the base station 160 and receives electromagnetic signals from the base station 160 using the one or more antennas 122a-n. The base station 160 communicates with the UE 102 using one or more antennas 180a-n.

It should be noted that in some configurations, one or more of the UEs 102 described herein may be implemented in a single device. For example, multiple UEs 102 may be combined into a single device in some implementations. Additionally or alternatively, in some configurations, one or more of the base stations 160 described herein may be implemented in a single device. For example, multiple base stations 160 may be combined into a single device in some implementations. In the context of FIG. 1, for instance, a single device may include one or more UEs 102 in accordance with the systems and methods described herein. Additionally or alternatively, one or more base stations 160

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in accordance with the systems and methods described herein may be implemented as a single device or multiple devices.

The UE **102** and the base station **160** may use one or more channels **119**, **121** to communicate with each other. For example, a UE **102** may transmit information or data to the base station **160** using one or more uplink (UL) channels **121** and signals. Examples of uplink channels **121** include a physical uplink control channel (PUCCH) and a physical uplink shared channel (PUSCH), etc. Examples of uplink signals include a demodulation reference signal (DMRS) and a sounding reference signal (SRS), etc. The one or more base stations **160** may also transmit information or data to the one or more UEs **102** using one or more downlink (DL) channels **119** and signals, for instance. Examples of downlink channels **119** include a PDCCH, a PDSCH, etc. A PDCCH can be used to schedule DL transmissions on PDSCH and UL transmissions on PUSCH, where the Downlink Control Information (DCI) on PDCCH includes downlink assignment and uplink scheduling grants. The PDCCH is used for transmitting Downlink Control Information (DCI) in a case of downlink radio communication (radio communication from the base station to the UE). Here, one or more DCIs (may be referred to as DCI formats) are defined for transmission of downlink control information. Information bits are mapped to one or more fields defined in a DCI format. Examples of downlink signals include a primary synchronization signal (PSS), a secondary synchronization signal (SSS), a cell-specific reference signal (CRS), a non-zero power channel state information reference signal (NZP CSI-RS), and a zero power channel state information reference signal (ZP CSI-RS), etc. Other kinds of channels or signals may be used.

Each of the one or more UEs **102** may include one or more transceivers **118**, one or more demodulators **114**, one or more decoders **108**, one or more encoders **150**, one or more modulators **154**, one or more data buffers **104** and one or more UE operations modules **124**. For example, one or more reception and/or transmission paths may be implemented in the UE **102**. For convenience, only a single transceiver **118**, decoder **108**, demodulator **114**, encoder **150** and modulator **154** are illustrated in the UE **102**, though multiple parallel elements (e.g., transceivers **118**, decoders **108**, demodulators **114**, encoders **150** and modulators **154**) may be implemented.

The transceiver **118** may include one or more receivers **120** and one or more transmitters **158**. The one or more receivers **120** may receive signals (e.g., downlink channels, downlink signals) from the base station **160** using one or more antennas **122a-n**. For example, the receiver **120** may receive and downconvert signals to produce one or more received signals **116**. The one or more received signals **116** may be provided to a demodulator **114**. The one or more transmitters **158** may transmit signals (e.g., uplink channels, uplink signals) to the base station **160** using one or more antennas **122a-n**. For example, the one or more transmitters **158** may upconvert and transmit one or more modulated signals **156**.

The demodulator **114** may demodulate the one or more received signals **116** to produce one or more demodulated signals **112**. The one or more demodulated signals **112** may be provided to the decoder **108**. The UE **102** may use the decoder **108** to decode signals. The decoder **108** may produce one or more decoded signals **106**, **110**. For example, a first UE-decoded signal **106** may comprise received payload data, which may be stored in a data buffer **104**. A second UE-decoded signal **110** may comprise overhead data and/or

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control data. For example, the second UE-decoded signal **110** may provide data that may be used by the UE operations module **124** to perform one or more operations.

As used herein, the term “module” may mean that a particular element or component may be implemented in hardware, software or a combination of hardware and software. However, it should be noted that any element denoted as a “module” herein may alternatively be implemented in hardware. For example, the UE operations module **124** may be implemented in hardware, software or a combination of both.

In general, the UE operations module **124** may enable the UE **102** to communicate with the one or more base stations **160**. The UE operations module **124** may include a UE RRC information configuration module **126**. The UE operations module **124** may include a UE resource management (RM) control module **128**. In some implementations, the UE operations module **124** may include physical (PHY) entities, Medium Access Control (MAC) entities, Radio Link Control (RLC) entities, packet data convergence protocol (PDCP) entities, and a Radio Resource Control (RRC) entity. For example, the UE RRC information configuration module **126** may process RRC parameter(s) included in the MIB and in the initial DL BWP configuration A. The UE RM control module **128** may determine whether the system information (e.g., SIB1) includes the initial DL BWP configuration B or not. The UE RM control module **128** may determine a bandwidth size C and a frequency position C for the CORESET #0 and determine a DL BWP C defined by the bandwidth size C and the frequency position C. The UE RM control module **128** may determine a bandwidth size A and a frequency position A for a DL BWP A from the initial DL BWP configuration A. The UE RM control module **128** may determine which of a DL BWP A and a DL BWP C is the initial DL BWP based on whether the bandwidth size A exceeds a predefined size. The UE RM control module **128** may determine an RB index of a first RB of a first RB group by using the frequency position A, which would be illustrated below.

The UE operations module **124** may provide information **148** to the one or more receivers **120**. For example, the UE operations module **124** may inform the receiver(s) **120** when or when not to receive transmissions based on the Radio Resource Control (RRC) message (e.g., broadcasted system information, RRC reconfiguration message), MAC control element, and/or the DCI (Downlink Control Information). The UE operations module **124** may provide information **148**, including the PDCCH monitoring occasions and DCI format size, to the one or more receivers **120**. The UE operation module **124** may inform the receiver(s) **120** when or where to receive/monitor the PDCCH candidate for DCI formats with which DCI size.

The UE operations module **124** may provide information **138** to the demodulator **114**. For example, the UE operations module **124** may inform the demodulator **114** of a modulation pattern anticipated for transmissions from the base station **160**.

The UE operations module **124** may provide information **136** to the decoder **108**.

For example, the UE operations module **124** may inform the decoder **108** of an anticipated encoding for transmissions from the base station **160**. For example, the UE operations module **124** may inform the decoder **108** of an anticipated PDCCH candidate encoding with which DCI size for transmissions from the base station **160**.

The UE operations module **124** may provide information **142** to the encoder **150**.

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The information **142** may include data to be encoded and/or instructions for encoding. For example, the UE operations module **124** may instruct the encoder **150** to encode transmission data **146** and/or other information **142**.

The encoder **150** may encode transmission data **146** and/or other information **142** provided by the UE operations module **124**. For example, encoding the data **146** and/or other information **142** may involve error detection and/or correction coding, mapping data to space, time and/or frequency resources for transmission, multiplexing, etc. The encoder **150** may provide encoded data **152** to the modulator **154**.

The UE operations module **124** may provide information **144** to the modulator **154**. For example, the UE operations module **124** may inform the modulator **154** of a modulation type (e.g., constellation mapping) to be used for transmissions to the base station **160**. The modulator **154** may modulate the encoded data **152** to provide one or more modulated signals **156** to the one or more transmitters **158**.

The UE operations module **124** may provide information **140** to the one or more transmitters **158**. This information **140** may include instructions for the one or more transmitters **158**. For example, the UE operations module **124** may instruct the one or more transmitters **158** when to transmit a signal to the base station **160**. The one or more transmitters **158** may upconvert and transmit the modulated signal(s) **156** to one or more base stations **160**.

The base station **160** may include one or more transceivers **176**, one or more demodulators **172**, one or more decoders **166**, one or more encoders **109**, one or more modulators **113**, one or more data buffers **162** and one or more base station operations modules **182**. For example, one or more reception and/or transmission paths may be implemented in a base station **160**. For convenience, only a single transceiver **176**, decoder **166**, demodulator **172**, encoder **109** and modulator **113** are illustrated in the base station **160**, though multiple parallel elements (e.g., transceivers **176**, decoders **166**, demodulators **172**, encoders **109** and modulators **113**) may be implemented.

The transceiver **176** may include one or more receivers **178** and one or more transmitters **117**. The one or more receivers **178** may receive signals (e.g., uplink channels, uplink signals) from the UE **102** using one or more antennas **180a-n**. For example, the receiver **178** may receive and downconvert signals to produce one or more received signals **174**. The one or more received signals **174** may be provided to a demodulator **172**. The one or more transmitters **117** may transmit signals (e.g., downlink channels, downlink signals) to the UE **102** using one or more antennas **180a-n**. For example, the one or more transmitters **117** may upconvert and transmit one or more modulated signals **115**.

The demodulator **172** may demodulate the one or more received signals **174** to produce one or more demodulated signals **170**. The one or more demodulated signals **170** may be provided to the decoder **166**. The base station **160** may use the decoder **166** to decode signals. The decoder **166** may produce one or more decoded signals **164**, **168**. For example, a first base station-decoded signal **164** may comprise received payload data, which may be stored in a data buffer **162**. A second base station-decoded signal **168** may comprise overhead data and/or control data. For example, the second base station-decoded signal **168** may provide data (e.g., PUSCH transmission data) that may be used by the base station operations module **182** to perform one or more operations.

In general, the base station operations module **182** may enable the base station **160** to communicate with the one or

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more UEs **102**. The base station operations module **182** may include a base station RRC information configuration module **194**. The base station operations module **182** may include a base station resource management (RM) control module **196** (or a base station RM processing module **196**). The base station operations module **182** may include PHY entities, MAC entities, RLC entities, PDCP entities, and an RRC entity.

The base station RM control module **196** may determine, for respective UE, different initial DL BWP configurations such as initial DL BWP configuration A and initial DL BWP configuration B and input the information to the base station RRC information configuration module **194**. The base station RM control module **196** may determine whether to generate system information to include both the initial DL BWP configuration A and the initial DL BWP configuration B or to generate system information to include the initial DL BWP configuration A but not the initial DL BWP configuration B. The base station RM control module **196** may determine whether an RRC parameter indicating a CSS set is generated in the initial DL BWP configuration B. The base station RM control module **196** may determine a bandwidth size C and a frequency position C for the CORESET #0 and generate a corresponding parameter in the MIB. The base station RM control module **196** may determine a bandwidth size A and a frequency position A for a DL BWP A and generate the initial DL BWP configuration A to indicate the bandwidth size A and the frequency position A. The base station RM control module **196** may determine which of a DL BWP A and a DL BWP C is the initial DL BWP for a UE based on whether the bandwidth size A exceeds a predefined size. The base station RM control module **196** may determine an RB index of a first RB of a first RB group by using the frequency position A, which would be illustrated below.

The base station operations module **182** may provide the benefit of performing PDCCH candidate search and monitoring efficiently. The base station operations module **182** may provide information **190** to the one or more receivers **178**. For example, the base station operations module **182** may inform the receiver(s) **178** when or when not to receive transmissions based on the RRC message (e.g., broadcasted system information, RRC reconfiguration message), MAC control element, and/or the DCI (Downlink Control Information).

The base station operations module **182** may provide information **188** to the demodulator **172**. For example, the base station operations module **182** may inform the demodulator **172** of a modulation pattern anticipated for transmissions from the UE(s) **102**.

The base station operations module **182** may provide information **186** to the decoder **166**. For example, the base station operations module **182** may inform the decoder **166** of an anticipated encoding for transmissions from the UE(s) **102**.

The base station operations module **182** may provide information **101** to the encoder **109**. The information **101** may include data to be encoded and/or instructions for encoding. For example, the base station operations module **182** may instruct the encoder **109** to encode transmission data **105** and/or other information **101**.

In general, the base station operations module **182** may enable the base station **160** to communicate with one or more network nodes (e.g., a NG mobility management function, a NG core UP functions, a mobility management entity (MME), serving gateway (S-GW), gNBs). The base station operations module **182** may also generate a RRC reconfiguration message to be signaled to the UE **102**.

The encoder 109 may encode transmission data 105 and/or other information 101 provided by the base station operations module 182. For example, encoding the data 105 and/or other information 101 may involve error detection and/or correction coding, mapping data to space, time and/or frequency resources for transmission, multiplexing, etc. The encoder 109 may provide encoded data 111 to the modulator 113. The transmission data 105 may include network data to be relayed to the UE 102.

The base station operations module 182 may provide information 103 to the modulator 113. This information 103 may include instructions for the modulator 113. For example, the base station operations module 182 may inform the modulator 113 of a modulation type (e.g., constellation mapping) to be used for transmissions to the UE(s) 102. The modulator 113 may modulate the encoded data 111 to provide one or more modulated signals 115 to the one or more transmitters 117.

The base station operations module 182 may provide information 192 to the one or more transmitters 117. This information 192 may include instructions for the one or more transmitters 117. For example, the base station operations module 182 may instruct the one or more transmitters 117 when to (or when not to) transmit a signal to the UE(s) 102. The base station operations module 182 may provide information 192, including the PDCCH monitoring occasions and DCI format size, to the one or more transmitters 117. The base station operation module 182 may inform the transmitter(s) 117 when or where to transmit the PDCCH candidate for DCI formats with which DCI size. The one or more transmitters 117 may upconvert and transmit the modulated signal(s) 115 to one or more UEs 102.

It should be noted that one or more of the elements or parts thereof included in the base station(s) 160 and UE(s) 102 may be implemented in hardware. For example, one or more of these elements or parts thereof may be implemented as a chip, circuitry or hardware components, etc. It should also be noted that one or more of the functions or methods described herein may be implemented in and/or performed using hardware. For example, one or more of the methods described herein may be implemented in and/or realized using a chipset, an application-specific integrated circuit (ASIC), a large-scale integrated circuit (LSI) or integrated circuit, etc.

A base station may generate a RRC message including the one or more RRC parameters, and transmit the RRC message to a UE. A UE may receive, from a base station, a RRC message including one or more RRC parameters. The term 'RRC parameter(s)' in the present disclosure may be alternatively referred to as 'RRC information element(s)'. A RRC parameter may further include one or more RRC parameter(s). In the present disclosure, a RRC message may include system information. A RRC message may include one or more RRC parameters. A RRC message may be sent on a broadcast control channel (BCCH) logical channel, a common control channel (CCCH) logical channel or a dedicated control channel (DCCH) logical channel.

In the present disclosure, a description 'a base station may configure a UE to' may also imply/refer to 'a base station may transmit, to a UE, an RRC message including one or more RRC parameters'. Additionally or alternatively, 'RRC parameter configure a UE to' may also refer to 'a base station may transmit, to a UE, an RRC message including one or more RRC parameters'. Additionally or alternatively, 'a UE is configured to' may also refer to 'a UE may receive, from a base station, an RRC message including one or more RRC parameters'.

FIG. 2 is a diagram illustrating one example of a resource grid 200.

For each numerology and carrier, a resource grid of $N_{grid,x}^{size,\mu} N_{sc}^{RB}$ subcarriers and $N_{symb}^{subframe,\mu}$ OFDM symbols is defined, starting at common resource block $N_{grid}^{start,\mu}$ indicated by higher layer signaling. There is one set of resource grids per transmission direction (uplink or downlink) with the subscript x set to DL and UL for downlink and uplink, respectively. There is one resource grid for a given antenna port p, subcarrier spacing configuration μ , and the transmission direction (downlink or uplink). When there is no risk for confusion, the subscript x may be dropped.

In the FIG. 2, the resource grid 200 includes the $N_{grid,x}^{size,\mu} N_{sc}^{RB}$ (202) subcarriers in the frequency domain and includes $N_{symb}^{subframe,\mu}$ (204) symbols in the time domain. In the FIG. 2, as an example for illustration, the subcarrier spacing configuration μ is set to 0. That is, in the FIG. 2, the number of consecutive OFDM symbols $N_{symb}^{subframe,\mu}$ (204) per subframe is equal to 14.

The carrier bandwidth $N_{grid}^{size,\mu}$ ($N_{grid,x}^{size,\mu}$) for subcarrier spacing configuration μ is given by the higher-layer (RRC) parameter carrierBandwidth in the SCS-SpecificCarrier IE. The starting position $N_{grid}^{start,\mu}$ for subcarrier spacing configuration μ is given by the higher-layer parameter offsetToCarrier in the SCS-SpecificCarrier IE. The frequency location of a subcarrier refers to the center frequency of that subcarrier.

In the FIG. 2, for example, a value of offset is provided by the higher-layer parameter offsetToCarrier. That is, $k=12 \times$ offset is the lowest usable subcarrier on this carrier.

Each element in the resource grid for antenna port p and subcarrier spacing configuration μ is called a resource element and is uniquely identified by $(k, l)_{p,\mu}$ where k is the index in the frequency domain and l refers to the symbols position in the time domain relative to same reference point. The resource element consists of one subcarrier during one OFDM symbol.

A resource block is defined as $N_{sc}^{RB}=12$ consecutive subcarriers in the frequency domain. As shown in the FIG. 2, a resource block 206 includes 12 consecutive subcarriers in the frequency domain. Resource block can be classified as common resource block (CRB) and physical resource block (PRB).

Common resource blocks are numbered from 0 and upwards in the frequency domain for subcarrier spacing configuration μ . The center of subcarrier 0 of common resource block with index 0 (i.e. CRB0) for subcarrier spacing configuration μ coincides with point A. The relation between the common resource block number n_{CRB}^{μ} in the frequency domain and resource element (k, l) for subcarrier spacing configuration μ is given by Formula (1) $n_{CRB}^{\mu} = \text{floor}((k/N_{sc}^{RB})^{\mu})$ where k is defined relative to the point A such that $k=0$ corresponds to the subcarrier centered around the point A. The function floor(A) hereinafter is to output a maximum integer not larger than the A.

Point A refers to as a common reference point. Point A coincides with subcarrier 0 (i.e. $k=0$) of a CRB 0 for all subcarrier spacing. Point A can be obtained from a RRC parameter offsetToPointA or a RRC parameter absoluteFrequencyPointA. The RRC parameter offsetToPointA is used for a PCell downlink and represents the frequency offset between point A and the lowest subcarrier of the lowest resource block, which has the subcarrier spacing provided by a higher-layer parameter subCarrierSpacingCommon and overlaps with the SS/PBCH block used by the UE for initial cell selection, expressed in units of resource blocks assum-

ing 15 kHz subcarrier spacing for frequency range (FR) 1 and 60 kHz subcarrier spacing for frequency range (FR2). FR1 corresponds to a frequency range between 410 MHz and 7125 MHz. FR2 corresponds to a frequency range between 24250 MHz and 52600 MHz. The RRC parameter absoluteFrequencyPointA is used for all cases other than the PCell case and represents the frequency-location of point A expressed as in ARFCN. The frequency location of point A can be the lowest subcarrier of the carrier bandwidth (or the actual carrier). Additionally, point A may be located outside the carrier bandwidth (or the actual carrier).

As above mentioned, the information element (IE) SCS-SpecificCarrier provides parameters determining the location and width of the carrier bandwidth or the actual carrier. That is, a carrier (or a carrier bandwidth, or an actual carrier) is determined (identified, or defined) at least by a RRC parameter offsetToCarrier, a RRC parameter subcarrierSpacing, and a RRC parameter carrierBandwidth in the SCS-SpecificCarrier IE.

The subcarrierSpacing indicates (or defines) a subcarrier spacing of the carrier. The offsetToCarrier indicates an offset in frequency domain between point A and a lowest usable subcarrier on this carrier in number of resource blocks (e.g. CRBs) using the subcarrier spacing defined for the carrier. The carrierBandwidth indicates width of this carrier in number of resource blocks (e.g. CRBs or PRBs) using the subcarrier spacing defined for the carrier. A carrier includes at most 275 resource blocks.

Physical resource block for subcarrier spacing configuration μ are defined within a bandwidth part and numbered from 0 to $N_{BWP,i}^{size,\mu}$ where i is the number of the bandwidth part. The relation between the physical resource block n_{PRB}^{μ} in bandwidth part (BWP) i and the common resource block n_{CRB}^{μ} is given by Formula (2) $n_{CRB}^{\mu} = n_{PRB}^{\mu} + N_{BWP,i}^{start,\mu}$ where $N_{BWP,i}^{start,\mu}$ is the common resource block where bandwidth part i starts relative to common resource block 0 (CRB0). When there is no risk for confusion the index μ may be dropped.

A BWP is a subset of contiguous common resource block for a given subcarrier spacing configuration μ on a given carrier. To be specific, a BWP can be identified (or defined) at least by a subcarrier spacing μ indicated by the RRC parameter subcarrierSpacing, a cyclic prefix determined by the RRC parameter cyclicPrefix, a frequency domain location, a bandwidth, an BWP index indicated by bwp-Id and so on. The locationAndBandwidth can be used to indicate the frequency domain location and bandwidth of a BWP. The value indicated by the locationAndBandwidth is interpreted as resource indicator value (RV) corresponding to an offset (a starting resource block) RB_{start} and a length L_{RB} in terms of contiguous resource blocks. The offset RB_{start} is a number of CRBs between the lowest CRB of the carrier and the lowest CRB of the BWP. The $N_{BWP,i}^{start,\mu}$ start, is given as Formula (3) $N_{BWP,i}^{start,\mu} = O_{carrier} + RB_{start}$. The value of $O_{carrier}$ is provided by offsetToCarrier for the corresponding subcarrier spacing configuration μ . In other words, the locationAndBandwidth may provide a starting RB index and a number of contiguous RBs for a BWP.

A UE 102 configured to operation in BWPs of a serving cell, is configured by higher layers for the serving cell a set of at most four BWPs in the downlink for reception. At a given time, a single downlink BWP is active. The base station 160 may not transmit, to the UE 102, PDSCH and/or PDCCH outside the active downlink BWP. A UE 102 configured to operation in BWPs of a serving cell, is configured by higher layers for the serving cell a set of at most four BWPs for transmission. At a given time, a single

uplink BWP is active. The UE 102 may not transmit, to the base station 160, PUSCH or PUCCH outside the active BWP. The specific signaling (higher layers signaling) for BWP configurations are described later.

FIG. 3 is a diagram illustrating one example 300 of common resource block grid, carrier configuration and BWP configuration by a UE 102 and a base station 160.

Point A 301 is a lowest subcarrier of a CRB0 for all subcarrier spacing configurations. The CRB grid 302 and the CRB grid 312 are corresponding to two different subcarrier spacing configurations. The CRB grid 302 is for subcarrier spacing configuration $\mu=0$ (i.e. the subcarrier spacing with 15 kHz). The CRB grid 312 is for subcarrier spacing configuration $\mu=1$ (i.e. the subcarrier spacing with 30 kHz).

One or more carriers are determined by respective SCS-SpecificCarrier IEs, respectively. In the FIG. 3, the carrier 304 uses the subcarrier spacing configuration $\mu=0$. And the carrier 314 uses the subcarrier spacing configuration $\mu=1$. The starting position $N_{grid}^{start,\mu}$ of the carrier 304 is given based on the value of an offset 303 (i.e. $O_{carrier}$) indicated by an offsetToCarrier in an SCS-SpecificCarrier IE. As shown in the FIG. 3, for example, the offsetToCarrier indicates the value of the offset 303 as $O_{carrier}=3$. That is, the starting position $N_{grid}^{start,\mu}$ of the carrier 304 corresponds to the CRB3 of the CRB grid 302 for subcarrier spacing configuration $\mu=0$. In the meantime, the starting position $N_{grid}^{start,\mu}$ of the carrier 314 is given based on the value of an offset 313 (i.e. $O_{carrier}$) indicated by an offsetToCarrier in another SCS-SpecificCarrier IE. For example, the offsetToCarrier indicates the value of the offset 313 as $O_{carrier}=1$. That is, the starting position $N_{grid}^{start,\mu}$ of the carrier 314 corresponds to the CRB1 of the CRB grid 312 for subcarrier spacing configuration $\mu=1$. A carrier using different subcarrier spacing configurations can occupy different frequency ranges.

As above-mentioned, a BWP is for a given subcarrier spacing configuration μ . One or more BWPs can be configured for a same subcarrier spacing configuration μ . For example, in the FIG. 3, the BWP 306 is identified at least by the $\mu=0$, a frequency domain location, a bandwidth (L_{RB}), and an BWP index (index A). The first PRB (i.e. PRB0) of a BWP is determined at least by the subcarrier spacing of the BWP, an offset derived by the locationAndBandwidth and an offset indicated by the offsetToCarrier corresponding to the subcarrier spacing of the BWP. An offset 305 (RB_{start}) is derived as 1 by the locationAndBandwidth. According to the Formulas (2) and (3), the PRB0 of BWP 306 corresponds to CRB 4 of the CRB grid 302, and the PRB1 of BWP 306 corresponds to CRB 5 of the CRB grid 302, and so on.

Additionally, in the FIG. 3, the BWP 308 is identified at least by the $\mu=0$, a frequency domain location, a bandwidth (L_{RB}), and an BWP index (index B). For example, an offset 307 (RB_{start}) is derived as 6 by the locationAndBandwidth. According to the Formulas (2) and (3), the PRB0 of BWP 308 corresponds to CRB 9 of the CRB grid 302, and the PRB1 of BWP 308 corresponds to CRB 10 of the CRB grid 302, and so on.

Additionally, in the FIG. 3, the BWP 316 is identified at least by the $\mu=1$, a frequency domain location, a bandwidth (L_{RB}), and an BWP index (index C). For example, an offset 315 (RB_{start}) is derived as 1 by the locationAndBandwidth. According to the Formulas (2) and (3), the PRB0 of BWP 316 corresponds to CRB 2 of the CRB grid 312, and the PRB1 of BWP 316 corresponds to CRB 3 of the CRB grid 312, and so on.

As shown in the FIG. 3, a carrier with the defined subcarrier spacing locate in a corresponding CRB grid with the same subcarrier spacing. A BWP with the defined

subcarrier spacing locate in a corresponding CRB grid with the same subcarrier spacing as well.

A base station may transmit a RRC message including one or more RRC parameters related to BWP configuration to a UE. A UE may receive the RRC message including one or more RRC parameters related to BWP configuration from a base station. For each cell, the base station may configure at least an initial DL BWP and one initial uplink bandwidth parts (initial UL BWP) to the UE. Furthermore, the base station may configure additional UL and DL BWPs to the UE for a cell.

A RRC parameters initialDownlinkBWP may indicate the initial downlink BWP (initial DL BWP) configuration for a serving cell (e.g., a SpCell and Scell). The base station may configure the RRC parameter locationAndBandwidth included in the initialDownlinkBWP so that the initial DL BWP contains the entire CORESET #0 of this serving cell in the frequency domain. The locationAndBandwidth may be used to indicate the frequency domain location and bandwidth of a BWP. A RRC parameters initialUplinkBWP may indicate the initial uplink BWP (initial UL BWP) configuration for a serving cell (e.g., a SpCell and Scell). The base station may transmit initialDownlinkBWP and/or initialUplinkBWP which may be included in SIB1, RRC parameter ServingCellConfigCommon, or RRC parameter ServingCellConfig to the UE.

The initialDownlinkBWP may include one, more or all of (I) generic parameters (e.g. locationAndBandwidth, subcarrierSpacing, cyclicPrefix) of the initial Downlink BWP, (II) cell specific parameters (e.g. pdccch-ConfigCommon) for PDCCH of the initial downlink BWP, (III) cell specific parameters (e.g. pdsch-ConfigCommon) for the PDSCH of the initial downlink BWP. The initialUplinkBWP may include one, more or all of (I) generic parameters (e.g. locationAndBandwidth, subcarrierSpacing, cyclicPrefix) of the initial UL BWP, (II) cell specific parameters (e.g. pucch-ConfigCommon) for PUCCH of the initial UL BWP, (III) cell specific parameters (e.g. pusch-ConfigCommon) for the PUSCH of the initial UL BWP, and (IV) cell specific random access parameters (e.g. rach-ConfigCommon).

SIB1, which is a cell-specific system information block (SystemInformationBlock, SIB), may contain information relevant when evaluating if a UE is allowed to access a cell and define the scheduling of other system information. SIB1 may also contain radio resource configuration information that is common for all UEs and barring information applied to the unified access control. The RRC parameter ServingCellConfigCommon is used to configure cell specific parameters of a UE's serving cell. The RRC parameter ServingCellConfig is used to configure (add or modify) the UE with a serving cell, which may be the SpCell or an Scell of an MCS or SCG. The RRC parameter ServingCellConfig herein are mostly UE specific but partly also cell specific.

The base station may configure the UE with a RRC parameter BWP-Downlink and a RRC parameter BWP-Uplink. The RRC parameter BWP-Downlink can be used to configure an additional DL BWP. The RRC parameter BWP-Uplink can be used to configure an additional UL BWP. The base station may transmit the BWP-Downlink and the BWP-Uplink which may be included in RRC parameter ServingCellConfig to the UE.

If a UE is not configured (provided) initialDownlinkBWP from a base station, an initial DL BWP is defined by a location and number of contiguous physical resource blocks (PRBs), starting from a PRB with the lowest index and ending at a PRB with the highest index among PRBs of a CORESET for Type0-PDCCH CSS set (i.e. CORESET #0),

and a subcarrier spacing (SCS) and a cyclic prefix for PDCCH reception in the CORESET for Type0-PDCCH CSS set. If a UE is configured (provided) initialDownlinkBWP from a base station, the initial DL BWP is provided by the initialDownlinkBWP. If a UE is configured (provided) initialUplinkBWP from a base station, the initial UL BWP is provided by the initialUplinkBWP.

The UE may be configured by the base station, at least one initial BWP and up to 4 additional BWP(s). One of the initial BWP and the configured additional BWP(s) may be activated as an active BWP. The UE may monitor DCI format, and/or receive PDSCH in the active DL BWP. The UE may not monitor DCI format, and/or receive PDSCH in a DL BWP other than the active DL BWP. The UE may transmit PUSCH and/or PUCCH in the active UL BWP. The UE may not transmit PUSCH and/or PUCCH in a BWP other than the active UL BWP.

As above-mentioned, a UE may monitor DCI format in the active DL BWP. To be more specific, a UE may monitor a set of PDCCH candidates in one or more CORESETs on the active DL BWP on each activated serving cell configured with PDCCH monitoring according to corresponding search space set where monitoring implies decoding each PDCCH candidate according to the monitored DCI formats.

A set of PDCCH candidates for a UE to monitor is defined in terms of PDCCH search space sets. A search space set can be a CSS set or a USS set. A UE may monitor a set of PDCCH candidates in one or more of the following search space sets

- a Type0-PDCCH CSS set configured by pdccch-ConfigSIB1 in MIB or by searchSpaceSIB1 in PDCCH-ConfigCommon or by searchSpaceZero in PDCCH-ConfigCommon for a DCI format with CRC scrambled by a SI-RNTI on the primary cell of the MCG
- a Type0A-PDCCH CSS set configured by searchSpaceOtherSystemInformation in PDCCH-ConfigCommon for a DCI format with CRC scrambled by a SI-RNTI on the primary cell of the MCG
- a Type1-PDCCH CSS set configured by ra-SearchSpace in PDCCH-ConfigCommon for a DCI format with CRC scrambled by a RA-RNTI or a TC-RNTI on the primary cell
- a Type2-PDCCH CSS set configured by paging-SearchSpace in PDCCH-ConfigCommon for a DCI format with CRC scrambled by a P-RNTI on the primary cell of the MCG
- a Type3-PDCCH CSS set configured by SearchSpace in PDCCH-Config with searchSpaceType=common for DCI formats with CRC scrambled by INT-RNTI, SFI-RNTI, TPC-PUSCH-RNTI, TPC-PUCCH-RNTI, or TPC-SRS-RNTI and, only for the primary cell, C-RNTI, MCS-C-RNTI, or CS-RNTI(s), and
- a USS set configured by SearchSpace in PDCCH-Config with searchSpaceType=ue-Specific for DCI formats with CRC scrambled by C-RNTI, MCS-C-RNTI, SP-CSI-RNTI, or C S-RNTI(s).

For a DL BWP, if a UE is configured (provided) one above-described search space set, the UE may determine PDCCH monitoring occasions for a set of PDCCH candidates of the configured search space set. PDCCH monitoring occasions for monitoring PDCCH candidates of a search space set *s* is determined according to the search space set *s* configuration and a CORESET configuration associated with the search space set *s*. In other words, a UE may monitor a set of PDCCH candidates of the search space set in the determined (configured) PDCCH monitoring occasions in one or more configured control resource sets

(CORESETs) according to the corresponding search space set configurations and CORESET configuration. A base station may transmit, to a UE, information to specify one or more CORESET configurations and/or one or more search space configurations. The information may be included in MIB and/or SIBs broadcasted by the base station. The information may be included in RRC configurations or RRC parameters. A base station may broadcast system information such as MIB, SIBs to indicate CORESET configuration(s) or search space configuration(s) to a UE. Or the base station may transmit a RRC message including one or more RRC parameters related to CORESET configuration(s) and/or search space configuration(s) to a UE.

An illustration of search space set configuration is described below.

A base station may transmit a RRC message including one or more RRC parameters related to search space configuration. A base station may determine one or more RRC parameter(s) related to search space configuration for a UE. A UE may receive, from a base station, a RRC message including one or more RRC parameters related to search space configuration. RRC parameter(s) related to search space configuration (e.g. SearchSpace, searchSpaceZero) defines how and where to search for PDCCH candidates. 'search/monitor for PDCCH candidate for a DCI format' may also refer to 'monitor/search for a DCI format' for short.

For example, a RRC parameter searchSpaceZero is used to configure a common search space 0 of an initial DL BWP. The searchSpaceZero corresponds to 4 bits. The base station may transmit the searchSpaceZero via PBCH(MIB) or ServingCell.

Additionally, a RRC parameter SearchSpace is used to define how/where to search for PDCCH candidates. The RRC parameters search space may include a plurality of RRC parameters as like, searchSpaceId, controlResourceSetId, monitoringSlotPeriodicityAndOffset, duration, monitoringSymbolsWithinSlot, nrofCandidates, searchSpaceType. Some of the above-mentioned RRC parameters may be present or absent in the RRC parameters SearchSpace. Namely, the RRC parameter SearchSpace may include all the above-mentioned RRC parameters. Namely, the RRC parameter SearchSpace may include one or more of the above-mentioned RRC parameters. If some of the parameters are absent in the RRC parameter SearchSpace, the UE may apply a default value for each of those parameters.

Herein, the RRC parameter searchSpaceId is an identity or an index of a search space. The RRC parameter searchSpaceId is used to identify a search space. Or rather, the RRC parameter searchSpaceId provide a search space set index s , $0 \leq s < 40$. Then a search space s hereinafter may refer to a search space identified by index s indicated by RRC parameter searchSpaceId. The above-mentioned searchSpaceSIB1, searchSpaceOtherSystemInformation, ra-SearchSpace, and pagingSearchSpace indicate respective index of a search space (i.e. searchSpaceId) so that corresponding search space configuration can be determined. Specifically, the searchSpaceSIB1 indicates an index for the Type0-PDCCH CSS set, i.e., search space ID, for SIB1 message. The searchSpaceOtherSystemInformation indicates an index for the Type0A-PDCCH CSS set for other system information, i.e., the system information other than MIB and SIB1. The ra-SearchSpace indicates an index for the Type1-PDCCH CSS set for random access procedure. The pagingSearchSpace indicates an index for the Type2-PDCCH CSS set for paging.

The RRC parameter controlResourceSetId concerns an identity of a CORESET, used to identify a CORESET. The

RRC parameter controlResourceSetId indicates an association between the search space s and the CORESET identified by controlResourceSetId. The RRC parameter controlResourceSetId indicates a CORESET applicable for the search space. CORESET p hereinafter may refer to a CORESET identified by index p indicated by RRC parameter controlResourceSetId. Each search space is associated with one CORESET. The RRC parameter monitoringSlotPeriodicityAndOffset is used to indicate slots for PDCCH monitoring configured as periodicity and offset. Specifically, the RRC parameter monitoringSlotPeriodicityAndOffset indicates a PDCCH monitoring periodicity of k_s slots and a PDCCH monitoring offset of o_s slots. A UE can determine which slot is configured for PDCCH monitoring according to the RRC parameter monitoringSlotPeriodicityAndOffset. The RRC parameter monitoringSymbolsWithinSlot is used to indicate a first symbol(s) for PDCCH monitoring in the slots configured for PDCCH monitoring. That is, the parameter monitoringSymbolsWithinSlot provides a PDCCH monitoring pattern within a slot, indicating first symbol(s) of the CORESET within a slot (configured slot) for PDCCH monitoring. The RRC parameter duration indicates a number of consecutive slots T_s that the search space lasts (or exists) in every occasion (PDCCH occasion, PDCCH monitoring occasion).

The RRC parameter may include aggregationLevel1, aggregationLevel2, aggregationLevel4, aggregationLevel8, aggregationLevel16. The RRC parameter nrofCandidates may provide a number of PDCCH candidates per CCE aggregation level L by aggregationLevel1, aggregationLevel2, aggregationLevel4, aggregationLevel8, and aggregationLevel16, for CCE aggregation level 1, CCE aggregation level 2, CCE aggregation level 4, for CCE aggregation level 8, and CCE aggregation level 16, respectively. In other words, the value L can be set to either one in the set $\{1, 2, 4, 8, 16\}$. The number of PDCCH candidates per CCE aggregation level L can be configured as 0, 1, 2, 3, 4, 5, 6, or 8. For example, in a case the number of PDCCH candidates per CCE aggregation level L is configured as 0, the UE may not search for PDCCH candidates for CCE aggregation L . That is, in this case, the UE may not monitor PDCCH candidates for CCE aggregation L of a search space set s . For example, the number of PDCCH candidates per CCE aggregation level L is configured as 4, the UE may monitor 4 PDCCH candidates for CCE aggregation level L of a search space set s .

The RRC parameter searchSpaceType is used to indicate that the search space set s is either a CSS set or a USS set. The RRC parameter searchSpaceType may include either a common or a ue-Specific. The RRC parameter common configure the search space set s as a CSS set and DCI format to monitor. The RRC parameter ue-Specific configures the search space set s as a USS set. The RRC parameter ue-Specific may include dci-Formats. The RRC parameter dci-Formats indicates to monitor PDCCH candidates either for DCI format 0_0 and DCI format 1_0, or for DCI format 0_1 and DCI format 1_1 in search space set s . That is, the RRC parameter searchSpaceType indicates whether the search space set s is a CSS set or a USS set as well as DCI formats to monitor for. The RRC parameter ue-Specific may further include a new RRC parameter (e.g. dci-FormatsExt) in addition to the dci-Formats. The RRC parameter dci-FormatsExt indicates to monitor PDCCH candidates for DCI format 0_2 and DCI format 1_2, or for DCI format 0_1, DCI format 1_1, DCI format 0_2 and DCI format 1_2. If the RRC parameter dci-FormatsExt is included in the RRC parameter ue-Specific, the UE may ignore the RRC parameter dci-

Formats. That is to say, the UE may not monitor the PDCCH candidates for DCI formats indicated by the RRC parameter dci-Format and may monitor the PDCCH candidates for DCI formats indicated by the RRC parameter dci-Format-sExt.

The UE 102 may monitor PDCCH candidates for DCI format 0_0 and/or DCI format 1_0 in either a CSS or a USS. The UE 102 may monitor PDCCH candidates for DCI format 0_1, DCI format 1_1, DCI format 0_2 and/or DCI format 1_2 only in a USS but cannot monitor PDCCH candidates for DCI format 0_1, DCI format 1_1, DCI format 0_2, and/or DCI format 1_2 in a CSS. The DCI format 0_1 may schedule up to two transport blocks for one PUSCH while the DCI format 0_2 may only schedule one transport block for one PUSCH. DCI format 0_2 may not consist of some fields (e.g., 'CBG transmission information' field), which may be present in DCI format 0_1. Similarly, the DCI format 1_1 may schedule up to two transport blocks for one PDSCH while the DCI format 1_2 may only schedule one transport block for one PDSCH. DCI format 1_2 may not consist of some fields (e.g., 'CBG transmission information' field), which may be present in DCI format 1_1. The DCI format 1_2 and DCI format 1_1 may consist of one or more same DCI fields (e.g., 'antenna port' field).

The base station 160 may schedule a UE 102 to receive PDSCH by a downlink control information (DCI). A DCI format provides DCI and includes one or more DCI fields. The one or more DCI fields in a DCI format are mapped to the information bits. As above-mentioned, the UE 102 can be configured by the base station 160 one or more search space sets to monitor PDCCH for detecting corresponding DCI formats. If the UE 102 detects a DCI format (e.g., the DCI format 1_0, the DCI format 1_1, or the DCI format 1_2) in a PDCCH, the UE 102 may be scheduled by the DCI format to receive a PDSCH.

A USS at CCE aggregation level L is defined by a set of PDCCH candidates for CCE aggregation L. A USS set may be constructed by a plurality of USS(s) corresponding to respective CCE aggregation level L. A USS set may consist of one or more USS(s) corresponding to respective CCE aggregation level L. Likewise, a CSS at CCE aggregation level L is defined by a set of PDCCH candidates for CCE aggregation L. A CSS set may be constructed by a plurality of CSS(s) corresponding to respective CCE aggregation level L. A CSS set may consist of one or more CSS(s) corresponding to respective CCE aggregation level L.

Herein, 'a UE monitor PDCCH for a search space set s' also refers to 'a UE may monitor a set of PDCCH candidates of the search space set s'. Alternatively, 'a UE monitor PDCCH for a search space set s' also refers to 'a UE may attempt to decode each PDCCH candidate of the search space set s according to the monitored DCI formats'. As above-mentioned, the PDCCH is used for transmitting or carrying Downlink Control Information (DCI). Thus, 'PDCCH', 'DCI format', and/or 'PDCCH candidate' are virtually interchangeable. In other words, 'a UE monitors PDCCH' implies 'a UE monitors PDCCH for a DCI format'. That is, 'a UE monitors PDCCH' implies 'a UE monitors PDCCH for detection of a configured DCI format'.

In the present disclosure, the term "PDCCH search space sets" may also refer to "PDCCH search space". A UE monitors PDCCH candidates in one or more search space sets. A search space sets can be a common search space (CSS) set or a UE-specific search space (USS) set. In some implementations, a CSS set may be shared/configured among multiple UEs. The multiple UEs may search PDCCH candidates in the CSS set. In some implementations, a USS

set is configured for a specific UE. The UE may search one or more PDCCH candidates in the USS set. In some implementations, a USS set may be at least derived from a value of C-RNTI addressed to a UE.

An illustration of CORESET configuration is described below.

A base station may configure a UE one or more CORESETs for each DL BWP in a serving cell. For example, a RRC parameter ControlResourceSetZero is used to configure CORESET #0 of an initial DL BWP. The RRC parameter ControlResourceSetZero corresponds to 4 bits. The base station may transmit ControlResourceSetZero, which may be included in MIB or RRC parameter ServingCellConfigCommon, to the UE. MIB may include the system information transmitted on BCH(PBCH). A RRC parameter related to initial DL BWP configuration may also include the RRC parameter ControlResourceSetZero. RRC parameter ServingCellConfigCommon is used to configure cell specific parameters of a UE's serving cell and contains parameters which a UE would typically acquire from SSB, MIB or SIBs when accessing the cell from IDLE. The CORESET #0 refers to a common CORESET with ID #0.

Additionally, a RRC parameter ControlResourceSet is used to configure a time and frequency CORESET other than CORESET #0. The RRC parameter ControlResourceSet may include a plurality of RRC parameters such as, ControlResourceSetId, frequencyDomainResource, duration, cce-REG-MappingType, precoderGranularity, tci-PresentInDCI, pdcch-DMRS-ScramblingID and so on.

Here, the RRC parameter ControlResourceSetId is an CORESET index p, used to identify a CORESET within a serving cell, where $0 < p < 12$. The RRC parameter duration indicates a number of consecutive symbols of the CORESET $N_{\text{sym}}^{\text{CORESET}}$ which can be configured as 1, 2 or 3 symbols. A CORESET consists of a set of $N_{\text{RB}}^{\text{CORESET}}$ resource blocks (RBs) in the frequency domain and $N_{\text{sym}}^{\text{CORESET}}$ symbols in the time domain. The RRC parameter frequencyDomainResource indicates the set of $N_{\text{RB}}^{\text{CORESET}}$ RBs for the CORESET. Each bit in the frequencyDomainResource corresponds a group of 6 consecutive RBs, with grouping starting from the first RB group in the BWP. The first (left-most/most significant) bit corresponds to the first RB group in the BWP, and so on. The first common RB of the first RB group has common RB index $6 \times \text{ceiling}(N_{\text{BWP}}^{\text{start}}/6)$. A bit that is set to 1 indicates that this RB group belongs to the frequency domain resource of this CORESET. Bits corresponding to a group of RBs not fully contained in the bandwidth part within which the CORESET is configured are set to zero. The ceiling(A) function hereinafter is to output a smallest integer not less than A.

According to the CORESET configuration, a CORESET (a CORESET #0 or a CORESET p) consists of a set of PRBs with a time duration of 1 to 3 OFDM symbols. The resource units Resource Element Groups (REGs) and Control Channel Elements (CCEs) are defined within a CORESET. A CCE consists of 6 REGs where a REG equals one resource block during one OFDM symbol. Control channels are formed by aggregation of CCE. That is, a PDCCH consists of one or more CCEs. Different code rates for the control channels are realized by aggregating different number of CCE. Interleaved and non-interleaved CCE-to-REG mapping are supported in a CORESET. Each resource element group carrying PDCCH carries its own DMRS.

FIG. 4 is a diagram illustrating one example of CORESET configuration in a BWP by a UE 102 and a base station 160.

FIG. 4 illustrates that a UE 102 is configured with three CORESETs for receiving PDCCH transmission in two BWPs. In the FIG. 4, 401 represent point A. 402 is an offset in frequency domain between point A 401 and a lowest usable subcarrier on the carrier 403 in number of CRBs, and the offset 402 is given by the offsetToCarrier in the SCS-SpecificCarrier IE. The BWP 405 with index A and the carrier 403 are for a same subcarrier spacing configuration μ . The offset 404 between the lowest CRB of the carrier and the lowest CRB of the BWP in number of RBs is given by the locationAndBandwidth included in the BWP configuration for BWP A. The BWP 407 with index B and the carrier 403 are for a same subcarrier spacing configuration μ . The offset 406 between the lowest CRB of the carrier and the lowest CRB of the BWP in number of RBs is given by the locationAndBandwidth included in the BWP configuration for BWP B.

For the BWP 405, two CORESETs are configured. As above-mentioned, a RRC parameter frequencyDomainResource in respective CORESET configuration indicates the frequency domain resource for respective CORESET. In the frequency domain, a CORESET is defined in multiples of RB groups and each RB group consists of 6 RBs. For example, in the FIG. 4, the RRC parameter frequencyDomainResource provides a bit string with a fixed size (e.g. 45 bits) as like '11010000 . . . 000000' for CORESET #1. That is, the first RB group, the second RB group, and the fourth RB group belong to the frequency domain resource of the CORESET #1. Additionally, the RRC parameter frequencyDomainResource provides a bit string with a fixed size (e.g. 45 bits) as like '00101110 . . . 000000' for CORESET #2. That is, the third RB group, the fifth RB group, the sixth RB group and the seventh RB group belong to the frequency domain resource of the CORESET #2.

For the BWP 407, one CORESET is configured. As above-mentioned, a RRC parameter frequencyDomainResource in the CORESET configuration indicates the frequency domain resource for the CORESET #3. In the frequency domain, a CORESET is defined in multiples of RB groups and each RB group consists of 6 RBs. For example, in the FIG. 4, the RRC parameter frequencyDomainResource provides a bit string with a fixed size (e.g. 45 bits) as like '11010000 . . . 000000' for CORESET #3. That is, the first RB group, the second RB group, and the fourth RB group belong to the frequency domain resource of the CORESET #3. Although the bit string configured for CORESET #3 is same as that for CORESET #1, the first RB group of the BWP B is different from that of the BWP A in the carrier. Therefore, the frequency domain resource of the CORESET #3 in the carrier is different from that of the CORESET #1 as well.

FIGS. 5A and 5B are diagrams illustrating one example 500-1 of REG numbering and one example 500-2 of CCE resource numbering for a CORESET.

The UE 102 may monitor a set of PDCCH candidates for a search space set in a CORESET p which consist of a set of $N_{RB}^{CORESET}$ PRBs and one sets of $N_{symb}^{CORESET}$ consecutive OFDM symbols. The resource blocks $N_{RB}^{CORESET}$ PRBs configured for the CORESET can be contiguous or can be not contiguous in the frequency domain. For the CORESET, the REGs within the CORESET are numbered in increasing order in time-first manner, starting with 0 for the first OFDM symbol and the lowest-numbered resource block in the CORESET. In FIG. 5A, REGs within the CORESET are numbered in increasing order in time-first manner, starting with 0 for the first OFDM symbol and the lowest-numbered resource block in the 502. The REGs

within the CORESET 502 are numbered by 0 to 35 by the time-first manner. The REGs for different PDCCH monitoring occasion in a same CORESET are numbered by the same way. That is, one or more PDCCH monitoring occasions in a same CORESET may have same REG mapping.

In FIG. 5B, $N_{CCE,p}$ is the number of CCEs, numbered from 0 to $(N_{CCE,p}-1)$, in the CORESET. The CORESET herein comprises of 6 CCEs. According to the CCE-to-REG mapping, UE 102 may determine a CCE comprising of which corresponding REGs. For non-interleaved CCE-to-REG mapping, all CCEs for a DCI with AL L are mapped in consecutive REG bundles of the CORESET. For example, for non-interleaved CCE-to-REG mapping, a CCE with index 0 (CCE #0) 506 comprises of 6 consecutive REGs with 0, 1, 2, 3, 4, 5. For interleaved CCE-to-REG mapping, REG bundles constituting the CCEs for a PDCCH are distributed in the frequency domain in units of REG bundles. A REG bundle i is defined as REGs $\{i*B, i*B+1, \dots, i*B+B-1\}$ where B is the REG bundle size indicated by the base station.

The UE 102 can determine the CCE indexes for aggregation level L corresponding to PDCCH candidates of a USS for a USS set based on the value of C-RNTI addressed to the UE. The UE 102 can determine the CCE indexes for aggregation level L corresponding to PDCCH candidates of a CSS for a CSS set without the value of C-RNTI addressed to the UE.

To be more specific, for a search space set s associated with CORESET p , the CCE indexes for aggregation level L corresponding to PDCCH candidate m_{s,n_CI} the search space set in slot n for an active DL BWP of a serving cell corresponding to carrier indicator field value, CIF value, n_CI are given by Formula (4) $L*((Y_{p,n} + \text{floor}((m_{s,n_CI} * N_{CCE,p}) / (L * M_{s,max}^{(L)})) + n_CI) \bmod (\text{floor}(N_{CCE,p} / L))) + i$. The parameters in the Formula (4) are illustrated as below: for any CSS, $Y_{p,n}$ is equal to 0, while for a USS, $Y_{p,n} = (A_p * Y_{p,n-1}) \bmod D$ where $Y_{p,-1} = n_{RNTI} \neq 0$, $A_p = 39827$ for $p \bmod 3 = 0$, $A_p = 39829$ for $p \bmod 3 = 1$, $A_p = 39839$ for $p \bmod 3 = 2$, and $D = 65537$; slot n can be denoted by n_{sf} representing the slot number within a radio frame with respect to the SCS configuration u ; $i = 0, \dots, L-1$; $N_{CCE,p}$ is the number of CCEs, numbered from 0 to $(N_{CCE,p}-1)$, in CORESET p ; n_{RNTI} is an value of C-RNTI provided by the base station for the UE; n_CI is the carrier indicator field value if the UE 102 is configured with a carrier indicator field for the serving cell on which PDCCH is monitored; otherwise, including for any CSS, the n_CI is equal to 0; $m_{s,n_CI} = 0, \dots, M_{s,n_CI}^{(L)} - 1$, where $M_{s,n_CI}^{(L)}$ is the number of PDCCH candidates the UE is configured to monitor for aggregation level L of the search space sets for a serving cell corresponding to n_CI ; for any CSS, $M_{s,max}^{(L)} = M_{s,0}^{(L)}$; for a USS, $M_{s,max}^{(L)}$ is the maximum of $M_{s,n_CI}^{(L)}$ over all configured n_CI values for a CCE aggregation level L of search space set s . m_{s,n_CI} is an index of a PDCCH candidate the UE configured to monitor per aggregation level L of the search space set s .

Here, in a CORESET associated with a search space set s , a set of CCEs for AL L are those determining CCE indexes where the PDCCH candidates, the UE 102 is configured to monitor for ALL of the search space set, are placed. Here, a set of CCEs for ALL can also refer to a USS. That is, a search space sets may comprise of one or more corresponding sets of CCEs for respective AL L . A set of CCEs can also refer to as 'a USS'. A set of CCEs for AL L can also refer to 'a USS at ALL'.

As above-mentioned, the UE 102 may receive, from the base station 160, a RRC message including one or more

RRC parameters related to search space configuration. The UE 102 may determine PDCCH monitoring occasions for PDCCH candidates for each search space set s based on the received RRC parameters. The UE 102 may monitor PDCCH candidates for each search space set s in the determined PDCCH monitoring occasions. For example, a RRC parameter (e.g. SearchSpace) may provide the UE 102 for a search space set s , that a PDCCH monitoring periodicity of k_s slots, a PDCCH monitoring offset of o_s slots, a duration of T_s , a PDCCH monitoring pattern within a slot, and so on.

In order to monitor a set of PDCCH candidates of a search space set, the UE may determine PDCCH monitoring occasions according to the search space set configuration and associated CORESET configuration. FIG. 6 is a diagram illustrating one example 600 how to determine PDCCH monitoring occasions for PDCCH candidates based on corresponding search space set configuration and CORESET configuration.

In FIG. 6, the PDCCH monitoring periodicity k_s is configured as 6 slots. The PDCCH monitoring offset o_s is configured as 2 slots. The duration T_s is configured as 2 slots. The subcarrier spacing configuration u is configured as 0, which means the subcarrier spacing of the active DL BWP is 15 kHz. In this case $u=0$, $N_{slot}^{frame,u}$ is equal to 10. That is, in a case $u=0$, the number of slots per frame is 10. n_{sf}^u is the slot number within a radio frame. That is, the value of n_{sf}^u is in a range of $\{0, \dots, N_{slot}^{frame,u}-1\}$.

The UE 102 may determine a PDCCH monitoring occasion on an active DL BWP from the PDCCH monitoring periodicity, the PDCCH monitoring offset, and the PDCCH monitoring pattern within a slot for each configured search space set s . For a search space set s , the UE 102, if the slot with number n_{sf}^u satisfies Formula (5) $(n_{sf}^u * N_{slot}^{frame,u} + n_{sf}^u - o_s) \bmod k_s = 0$, may determine that a PDCCH monitoring occasion(s) exists in a slot with number n_{sf}^u in a frame with number n_f . According to Formula (5), the UE 102 may determine the slots with number $n_{sf}^u=2$ and $n_{sf}^u=8$ in a frame with number $n_f=0$ and the slot with number $n_{sf}^u=4$ in a frame with number $n_f=1$ as the slots in which the PDCCH monitoring occasions exists. Given the T_s is configured as 2 slots, the UE 102 may monitor PDCCH candidates for search space set s for $T_s=2$ consecutive slots, starting from the determined the slots with number n_{sf}^u . In other words, the UE 102 may not monitor PDCCH candidates for search space set s for the next $(k_s - T_s)$ consecutive slots. As depicted in FIG. 6, the UE 102 may determine the slots with number $n_{sf}^u=2, 3, 8$, and 9 in a frame with number $n_f=0$ and the slots with number $n_{sf}^u=4$, and 5 in a frame with number $n_f=1$ as the slots having PDCCH monitoring occasions. The UE 102 may monitor PDCCH candidates for search space set s in the determined slots configured for PDCCH monitoring. A slot having PDCCH monitoring occasions may also refer to a slot configured for PDCCH monitoring.

Furthermore, a slot determined (or configured) for PDCCH monitoring may have one or more than one PDCCH monitoring occasions. PDCCH monitoring pattern within the slot configured for PDCCH monitoring is indicated by a 14-bits string (monitoringSymbolsWithinSlot). Each bit within the 14-bits string may correspond to a symbol within a slot, respectively. The most significant (left) bit (MSB) may represent the first OFDM in a slot, and the second most significant (left) bit may represent the second OFDM symbol in a slot and so on. The bit(s) set to one may identify the first OFDM symbol(s) of the control resource set within a slot. As depicted in FIG. 6, a slot 602 configured for PDCCH monitoring may have two PDCCH monitoring

occasions. The first PDCCH monitoring occasion 604 is located on the first, second and third consecutive symbols. The second PDCCH monitoring occasion 606 is located on the 8th, 9th, and 10th consecutive OFDM symbols. The duration of one PDCCH monitoring occasion is equal to the duration of a CORESET associated with the search space set s . Generally, the duration of one PDCCH monitoring occasion (the number of the consecutive OFDM symbols for one PDCCH monitoring occasion) can be 1, 2 or 3 symbols. In the FIG. 6, a CORESET comprises one PDCCH monitoring occasion with 3 consecutive OFDM symbols in the time domain.

According to the FIG. 6, the UE may monitor a set of PDCCH candidates for the search space set s in the first PDCCH monitoring occasion 604 in the associated CORESET and may further monitor a set of PDCCH candidates for the search space set s in the second PDCCH monitoring occasion 606 in the CORESET in each slot in which the PDCCH monitoring is configured for the search space set s . Here, each PDCCH candidate for the search space set s is mapped in a resource of the associated CORESET in each PDCCH monitoring occasion. In other words, one PDCCH candidate for the search space set s is mapped to one associated CORESET in one PDCCH monitoring occasion. One PDCCH candidate for the search space set s is not mapped to more than one associated CORESET in different PDCCH monitoring occasions. For example, one PDCCH candidate for the search space set s is not mapped to both the first PDCCH monitoring occasion 604 and the second PDCCH monitoring occasion 606.

In the present disclosure, the UE 102 may have three RRC states (RRC modes), i.e., RRC_IDLE, RRC_INACTIVE, RRC_CONNECTED. RRC states can be also called as mobility states. Three RRC states can be applied to the UE 102, which is helpful to efficiently manage the power consumption of the UE 102. Although the UE 102 may have three RRC state, the UE 102 has only one RRC state at a given time, that is the UE 102 is in one RRC state at a given time. The UE 102 in RRC_CONNECTED state may need to constantly monitor control channel for data transmission and reception. Therefore, unless there is data required to be transmitted between the UE 102 and the base station 160, it is not necessary for the UE 102 to be always in RRC_CONNECTED state. In this case, the UE 102 can move from the RRC_CONNECTED state to either RRC_IDLE state or RRC_INACTIVE to at least save some power consumption.

To establish a RRC connection with a base station (or network), the UE 102 may initiate a random access procedure. When the RRC connection has been established, the UE 102 is either in RRC_CONNECTED state or in RRC_INACTIVE state. Moreover, when the RRC connection has been established but suspended, the UE 102 is in RRC_INACTIVE state. When the RRC connection has been established but not suspended, the UE 102 is in RRC_CONNECTED state. On the contrary, if no RRC connection is established, the UE 102 is in RRC_IDLE state.

In NR Release 15/16, the maximum bandwidth that NR Release 15/16 UEs (i.e. legacy UEs) can support are up to 100 MHz for FR1 and 200 MHz for FR2. Compared with the Release 15/16 UEs, cost reduction for a new UE type (e.g., wearable devices, industrial sensors, video surveillance) is desirable. To reduce the cost and the complexity, the UE with new type would be equipped with less reception antennas and/or the reduced bandwidth (i.e. RF bandwidth and/or baseband bandwidth) relative to the NR Release 15/16 UEs. The reduced reception antennas would result in a reduced power for the received channels/signals. The

reduced bandwidth would also result in a reduced frequency diversity. The maximum bandwidth that UEs with reduced bandwidth can support may be, for example, 20 MHz for FR1 and 100 MHz for FR2. This kind of UEs can be termed ‘RedCap (reduced capability) UEs’. The NR Release 15/16 UEs can be termed ‘non-RedCap UEs’. Additionally, UEs other than RedCap UEs can be termed ‘non-RedCap UEs’ as well. Unless otherwise specified, a UE **102** hereinafter in the present disclosure may refer to the RedCap UEs with reduced capabilities including reduced RF bandwidth and/or reduced baseband bandwidth. That is, the maximum bandwidth the UE **102** can support may be 20 MHz for FR1 and may be 100 MHz for FR2.

In a serving cell, the base station may configure BWPs (DL BWPs and/or UL BWPs) with different bandwidths and different frequency locations for different UEs. For a UE, a configurable bandwidth of a BWP (a DL BWP and/or a UL BWP) is subject to the UE’s bandwidth capability, i.e. the maximum bandwidth the UE can support. The base station may not configure a UE with a BWP whose bandwidth is wider than the maximum bandwidth the UE can support. The UE may not operate with a BWP whose bandwidth is wider than the maximum bandwidth the UE can support. In a serving cell, due to different bandwidth capabilities of different UEs, for example, the base station may configure a non-RedCap UE with a BWP whose bandwidth can be up to 100 MHz and may configure a RedCap UE with a BWP whose bandwidth can be up to 20 MHz.

In the present disclosure, as above-mentioned, the RRC parameter `initialDownlinkBWP` may indicate the initial downlink BWP configuration for a serving cell (e.g., a SpCell and Scell) while the RRC parameter `initialUplinkBWP` may indicate the initial UL BWP configuration for a serving cell (e.g., a SpCell and Scell). The base station may transmit `initialDownlinkBWP` and/or `initialUplinkBWP` which may be included in SIB1, an RRC parameter `ServingCellConfigCommon`, or RRC parameter `ServingCellConfig` to the UE. The RRC parameters `initialDownlinkBWP` included in the SIB1 is used to indicate the initial downlink BWP configuration for a primary cell. The RRC parameters `initialUplinkBWP` included in the SIB1 is used to indicate the initial UL BWP configuration for a primary cell.

In the present disclosure, the base station may allow various UEs (e.g., RedCap UEs and non-RedCap UEs) to camp on a serving cell. The base station may configure one initial DL BWP configuration (i.e., initial DL BWP configuration A) or two initial DL BWP configurations (i.e., initial DL BWP configuration A and initial DL BWP configuration B) for the serving cell according to real deployment. The base station may transmit, to these UEs, system information (e.g., SIB1) which includes the initial DL BWP configuration A or includes both the initial DL BWP configuration A and the initial DL BWP configuration B. A UE can receive, from the base station, the system information which includes the initial DL BWP configuration A or includes both the initial DL BWP configuration A and the initial DL BWP configuration B. The system information herein may mean MIB, SIB1, or other SIBs.

For a serving cell where both the non-RedCap UEs and RedCap UEs are allowed to camp, the base station may configure the non-RedCap UEs and RedCap UEs with the initial DL BWP configuration A. That is, the base station may not configure the RedCap UEs with the initial DL BWP configuration B for the serving cell. Additionally or alternatively, the base station may configure the non-RedCap UEs with the initial DL BWP configuration A and may

configure the RedCap UEs with the initial DL BWP configuration B for the serving cell. In the present disclosure, even if the system information includes the initial DL BWP configuration B (i.e., the RedCap UEs is additionally configured with the initial DL BWP configuration B), the RedCap UEs may also apply some part of the initial DL BWP configuration A for communication with the base station. Additionally, even if the system information includes the initial DL BWP configuration B, the non-RedCap UEs may not apply the initial DL BWP configuration B for communication with the base station.

The system information may include the initial DL BWP configuration A which is indicated by the RRC parameter `initialDownlinkBWP`. The initial DL BWP configuration A provides the UE **102** an initial DL BWP A. To be specific, the RRC parameter `initialDownlinkBWP` is included in an RRC parameter `DownlinkConfigCommonSIB` which provides common downlink parameters of a serving cell. The `DownlinkConfigCommonSIB` is included in the system information (e.g., the SIB1). In other words, in the present disclosure, the initial DL BWP configuration A may be referred to as the `initialDownlinkBWP` included in the `DownlinkConfigCommonSIB`. For convenience of illustration, ‘initial DL BWP A’ hereinafter may refer to ‘DL BWP A’. That is, in the present disclosure, ‘initial DL BWP A’ and ‘DL BWP A’ can be interchanged with each other.

On the other hand, the system information may include the initial DL BWP configuration B which is indicated by the RRC parameter `initialDownlinkBWP-redcap`. The initial DL BWP configuration B provides the UE **102** an initial DL BWP B. The RRC parameter `initialDownlinkBWP-redcap` can be included in the RRC parameter `DownlinkConfigCommonSIB`. Additionally or alternatively, the RRC parameter `initialDownlinkBWP-redcap` can be included in a RRC parameter `DownlinkConfigCommonRedCapSIB` which provides common downlink parameters of a serving cell specific to RedCap UEs. The `DownlinkConfigCommonRedCapSIB` is included in the system information (e.g., the SIB1) as well. In the present disclosure, the initial DL BWP configuration B may be referred to as the `initialDownlinkBWP-redcap` included in the `DownlinkConfigCommonSIB`. Additionally or alternatively, the initial DL BWP configuration B may be referred to as the `initialDownlinkBWP` included in the `DownlinkConfigCommonRedCapSIB`. In other words, the `initialDownlinkBWP-redcap` hereinafter may also refer to the `initialDownlinkBWP` included in the `DownlinkConfigCommonRedCapSIB`. Unless otherwise specified, the `initialDownlinkBWP` hereinafter refers to the `initialDownlinkBWP` included in the `DownlinkConfigCommonSIB`, and the `initialDownlinkBWP-redcap` hereinafter refers to the `initialDownlinkBWP` (or the `initialDownlinkBWP-redcap`) included in the `DownlinkConfigCommonRedCapSIB`. For convenience of illustration, ‘initial DL BWP B’ hereinafter may refer to ‘DL BWP B’. That is, in the present disclosure, ‘initial DL BWP B’ and ‘DL BWP B’ can be interchanged with each other.

In the present disclosure, the `initialDownlinkBWP-redcap` is used to indicate the initial Downlink BWP configuration B for a primary cell which is specific to RedCap UEs. On the other hand, the `initialDownlinkBWP` is used to indicate the initial Downlink BWP configuration A for a primary cell which can be applied to both the non-RedCap UEs and the RedCap UEs. The `initialDownlinkBWP-redcap` may have same information element structure as `initialDownlinkBWP`. That is, the `initialDownlinkBWP-redcap` may include some or all of RRC parameters included in

the initialDownlinkBWP. For example, the initialDownlinkBWP-redCap may also include generic parameters (e.g. locationAndBandwidth, subcarrierSpacing, cyclicPrefix) of the initial Downlink BWP, cell specific parameters (e.g. pdcch-ConfigCommon) for PDCCH of the initial downlink BWP, cell specific parameters (e.g. pdsch-ConfigCommon) for the PDSCH of the initial downlink BWP.

As above-mentioned, locationAndBandwidth is used to indicate the frequency domain location and bandwidth of a BWP. The base station may configure the locationAndBandwidth included in the initialDownlinkBWP (i.e., the initial DL BWP configuration A) so that the DL BWP A contains the entire CORESET #0 of the serving cell in the frequency domain. That is, the DL BWP A provided by the initial DL BWP configuration A shall contain the entire CORESET #0 in the frequency domain. However, the DL BWP B provided by the initial DL BWP configuration B may contain the entire CORESET #0 in the frequency domain or may not contain the entire CORESET #0 in the frequency domain. In other words, the base station may configure the locationAndBandwidth included in the initialDownlinkBWP-redCap (i.e., the initial DL BWP configuration B) so that the DL BWP B contains or does not the entire CORESET #0 of the serving cell in the frequency domain. To be specific, 'a DL BWP contains the (entire) CORESET #0 in the frequency domain' means 'the frequency domain location and bandwidth of the CORESET #0 are confined within a DL BWP' or means 'a DL BWP includes all RBs of the CORESET #0'. 'a DL BWP does not contain the (entire) CORESET #0 in the frequency domain' means 'the frequency domain location and bandwidth of the CORESET #0 are not confined within a DL BWP' or means 'a DL BWP does not include all RBs of the CORESET #0, i.e., a DL BWP includes zero or some RBs of the CORESET #0'. The frequency domain location and bandwidth of the DL BWP A can be different from those of the DL BWP B. The bandwidth of the DL BWP A can be larger than 20 MHz, i.e., the maximum bandwidth of the RedCap UE can support. While the bandwidth of the DL BWP B cannot be wider than 20 MHz.

In the present disclosure, for operation on the primary cell, the base station 160 may configure an initial Downlink BWP for non-RedCap UEs and configure another initial Downlink BWP for RedCap UEs separately from the initial Downlink BWP for non-RedCap UEs. RedCap UEs and non-RedCap UEs may not share a same initial Downlink BWP. Additionally or alternatively, for operation on the primary cell, the base station 160 may configure a same initial Downlink BWP for both non-RedCap UEs and RedCap UEs.

In the present disclosure, for both the RedCap UEs and the non-RedCap UEs, before these UEs receive the system information which includes the initial DL BWP configuration A and/or the initial DL BWP configuration B, i.e., these UEs are not provided the initial DL BWP configuration A and/or the initial DL BWP configuration B, these UEs may determine that an initial DL BWP is defined by a location and number of contiguous PRBs, starting from a PRB with the lowest index and ending at a PRB with the highest index among PRBs of the CORESET #0, and a SCS and a cyclic prefix for PDCCH reception in the CORESET #0. The CORESET #0 is also referred to as a CORESET for Type0-PDCCH CSS set. The SCS for PDCCH reception in the CORESET #0 is indicated by a RRC parameter included in the MIB. The cyclic prefix for PDCCH reception in the CORESET #0 is determined by the UE 102 and the base station 160 as normal cyclic prefix. That is, the UE 102 and the base station 160 may not determine the cyclic prefix as

the extended cyclic prefix. The UE 102 and the base station 160 may use the normal cyclic prefix for the initial DL BWP.

For non-RedCap UEs, upon the reception of the system information (e.g., the SIB1), the UEs are provided the initialDownlinkBWP, i.e. the initial DL BWP configuration A. If the UEs are provided the initial DL BWP configuration A, the UEs may determine that the initial DL BWP is the DL BWP provided by the initial DL BWP configuration A (i.e. the DL BWP A), regardless of whether the system information includes the initial DL BWP configuration B. As above-mentioned, the locationAndBandwidth included in the initial DL BWP configuration A provides a (total) number of contiguous RBs, L_{RB} , and an RB offset, RB_{start} . A starting RB index N_{BWP}^{start} is calculated by using the RB offset (RB_{start}) according to $N_{BWP,i}^{start} = O_{carrier} + RB_{start}$. The starting RB index N_{BWP}^{start} means an RB index of a common RB where the DL BWP A starts relative to CRB 0 on the CRB grid. A common RB where the DL BWP A starts represents a starting or first RB of the DL BWP A. The starting RB index N_{BWP}^{start} means an RB index of a starting (first) RB of the DL BWP A relative to CRB 0 on the CRB grid. The (total) number of contiguous RBs, L_{RB} , corresponds to the frequency bandwidth of the DL BWP A. In the present disclosure, the frequency position (location) for a DL BWP may also means a frequency position of a starting (first) RB of the DL BWP. A frequency position of a starting (first) RB of the DL BWP can be represented by an RB index of starting (first) RB of the DL BWP.

In other words, the UE may determine a (total) number of contiguous RBs and a starting RB index by using the locationAndBandwidth included in the initial DL BWP configuration A. That is, the locationAndBandwidth included in the initial DL BWP configuration A provides a total number (or quantity) of contiguous RBs and a starting RB index. For convenience of illustration, a total number (or quantity) of contiguous RBs provided by the locationAndBandwidth which is included in the initial DL BWP configuration A can be referred to as a total number (or quantity) A of contiguous RBs. Likewise, a starting RB index provided by the locationAndBandwidth which is included in the initial DL BWP configuration A can be referred to as a starting RB index A. The UE 102 and the base station 160 may determine the frequency location and bandwidth of the initial DL BWP by using the starting RB index A and the total number A of contiguous RBs. Additionally, the base station may determine a SCS for the initial DL BWP and generate, in the initial DL BWP configuration A, an RRC parameter subcarrierSpacing indicating the determined SCS. The UE may determine the SCS for the initial DL BWP according to the RRC parameter subcarrierSpacing included in the initial DL BWP configuration A. Additionally, the base station may determine to use normal cyclic prefix or extended cyclic prefix for the initial DL BWP. If the base station determines to use normal cyclic prefix for the initial DL BWP, the base station may not generate an RRC parameter cyclicPrefix in the initial DL BWP configuration A. On the other hand, if the base station determines to use extended cyclic prefix for the initial DL BWP, the base station may generate the RRC parameter cyclicPrefix in the initial DL BWP configuration A. The UE may determine to use normal cyclic prefix or extended cyclic prefix for the initial DL BWP based on whether the RRC parameter cyclicPrefix is present or not. The base station may transmit, to the UE, PDCCH or PDSCH in the initial DL BWP according to the SCS and the cyclic prefix (the determined CP length) determined for the initial DL BWP.

In one example A of the present disclosure, the base station may transmit system information (e.g. SIB1) which includes both the initial DL BWP configuration A and the initial DL BWP configuration B. In this case, for RedCap UEs, upon the reception of the system information (e.g., the SIB1), the UEs **102** are provided the initialDownlinkBWP-redCap, i.e. the initial DL BWP configuration B. If the UE **102** is provided the initial DL BWP configuration B, the UE **102** may determine that the initial DL BWP is the DL BWP provided by the initial DL BWP configuration B (i.e. the DL BWP B) and may not determine that the initial DL BWP is the DL BWP provided by the initial DL BWP configuration A (i.e. the DL BWP A). To be specific, the UE **102** may use the locationAndBandwidth included in the initial DL BWP configuration B to determine the frequency domain location (position) and the bandwidth of the DL BWP provided by the initial DL BWP configuration B. The locationAndBandwidth included in the initial DL BWP configuration B provides a (total) number of contiguous RBs and an RB offset RB_{start} . Then the starting RB index N_{BWP}^{start} of the DL BWP B can be calculated according to $N_{BWP}^{start} = O_{carrier} + RB_{start}$. That is, the starting RB index N_{BWP}^{start} means an RB index of a common RB where the DL BWP B starts relative to CRB 0 on the CRB grid. For convenience of illustration, a total number (or quantity) of contiguous RBs provided by the locationAndBandwidth which is included in the initial DL BWP configuration B can be referred to as a total number B (or quantity) of contiguous RBs. Likewise, a starting RB index provided (or determined) by the locationAndBandwidth which is included in the initial DL BWP configuration B can be referred to as a starting RB index B. Then the UE **102** and the base station **160** may determine the frequency location and bandwidth of the initial DL BWP by using the starting RB index B and the total number B of contiguous RBs.

In the example A, additionally, the base station may determine a SCS for the initial DL BWP and generate, in the initial DL BWP configuration B, an RRC parameter subcarrierSpacing indicating the determined SCS. The UE may determine the SCS for the initial DL BWP according to the RRC parameter subcarrierSpacing included in the initial DL BWP configuration B.

In the example A, additionally, the base station may determine to use normal cyclic prefix or extended cyclic prefix for the initial DL BWP. If base station determines to use normal cyclic prefix for the initial DL BWP, the base station may not generate an RRC parameter cyclicPrefix in the initial DL BWP configuration B. On the other hand, if the base station determines to use extended cyclic prefix for the initial DL BWP, the base station may generate the RRC parameter cyclicPrefix in the initial DL BWP configuration B. The UE **102** may determine to use normal cyclic prefix or extended cyclic prefix for the initial DL BWP based on whether the RRC parameter cyclicPrefix is present or not. The base station may transmit, to the UE **102**, PDCCH or PDSCH in the initial DL BWP according to the SCS and the cyclic prefix (the determined CP length) determined for the initial DL BWP.

In the present disclosure, for a serving cell where both the non-RedCap UEs and RedCap UEs are allowed by the base station **160** to camp, the base station **160** may make RedCap UEs use some part or all of the initial DL BWP configuration A such that the signaling overhead could be reduced. Then the base station may generate the initial DL BWP configuration A in the system information (e.g. SIB1) and may not generate the initial DL BWP configuration B in the system information (e.g. SIB1). The base station may broadcast or

transmit the system information (e.g. SIB1) which includes the initial DL BWP configuration A and does not include the initial DL BWP configuration B. As well as the non-RedCap UEs, the RedCap UEs may camp on the serving cell and may receive the system information which includes the initial DL BWP configuration A and does not include the initial DL BWP configuration B.

In this case where the UE **102** receives the system information which includes the initial DL BWP configuration A and does not include the initial DL BWP configuration B, illustrations of how the UE **102** determines the initial DL BWP are described below.

FIG. 8 is a flow diagram illustrating one implementation of a method **800** for initial DL BWP determination by a UE **102**.

In the implementation **800** of the present disclosure, the base station **160** may generate system information (e.g., SIB1) to at least include the initial DL BWP configuration A and not include the initial DL BWP configuration B.

The base station **160** may broadcast or transmit **801**, to the UE **102**, a set of SS/PBCH blocks in the serving cell. The UE **102** may receive **801**, from the base station **160**, the MIB. The UE **102** may blindly detect a set of SS/PBCH blocks in the frequency domain in the serving cell. Upon detection of a SS/PBCH block with an index, the UE **102** may acquire or receive the MIB from the SS/PBCH block.

Moreover, upon reception of the MIB, the UE **102** may determine, from the MIB, frequency position and bandwidth of a CORESET for Type0-PDCCH CSS set (i.e., CORESET #0) and the Type0-PDCCH CSS set. The MIB includes a configuration of the CORESET #0, i.e. controlResourceSetZero. The configuration of the CORESET #0 provides a total number of contiguous RBs, a number of contiguous symbols, and an RB offset for the CORESET #0. The UE **102** may determine the frequency position of a starting (first) RB of the CORESET #0 by using the RB offset wherein the RB offset is defined with respect to the SCS of the CORESET #0 and is an offset from the smallest RB index of the CORESET #0 to the smallest RB index of the common RB overlapping with the first RB of the detected SS/PBCH block. MIB provides the SCS of the CORESET #0. Furthermore, MIB also provides a k_{ssb} offset in number of subcarriers between the SS/PBCH block and the overall common RB grid. To be specific, the k_{ssb} offset in number of subcarriers is between the lowest subcarrier (#0) of a first RB of the SS/PBCH block and a subcarrier #0 of a common RB with smallest RB index overlapping with the first RB of the SS/PBCH block. Then the UE **102** determines the frequency location for the smallest RB index of the common RB overlapping with a first RB of the detected SS/PBCH block according to the k_{ssb} offset provided by the MIB.

In other words, the configuration of the CORESET #0 and/or the MIB provides the UE **102** a total number of contiguous RBs, a number of contiguous symbols, and a frequency position of a starting (first) RB for the CORESET #0. The base station **160** may firstly determine information such as a total number of contiguous RBs, a number of contiguous symbols, and a frequency position of a starting (first) RB for the CORESET #0 and then generate MIB to indicate the information. The UE **102** may determine **801**, from the received MIB, a total number of contiguous RBs, a number of contiguous symbols, and a frequency position of a starting (first) RB for the CORESET #0.

For convenience of illustration, a total number (or quantity) of contiguous RBs provided by the configuration of the CORESET #0 can be referred to as a total number (or quantity) C of contiguous RBs. That is, the total number C

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is the size of the CORESET #0 in number of RBs or is the frequency bandwidth for the CORESET #0 in units of RBs. Likewise, a frequency position of a starting (first) RB of the CORESET #0 can be referred to as a starting RB index C, an RB index C, or a frequency position C. 'a starting RB index C', 'an RB index C', and 'frequency position C' can be interchanged with each other. For illustration purpose, an RB index C may mean or represent the frequency position of the first RB of the CORESET #0 in the frequency domain.

The base station 160 may transmit 802, to the UE 102, a PDCCH for the Type 0-PDCCH CSS set and system information (e.g., SIB1) in a PDSCH wherein the PDSCH is scheduled by the PDCCH. The UE 102 may monitor 802 a PDCCH according to the Type 0-PDCCH CSS set. Upon detection of the PDCCH, the UE 102 may receive the SIB1 in a PDSCH scheduled by the PDCCH.

Upon reception of the SIB1, the UE 102 is provided the initial DL BWP configuration A. As above-mentioned, before the UE 102 is provided the initial DL BWP configuration A, the UE 102 may determine that the initial DL BWP of the UE 102 is defined by a location and number of contiguous PRBs, starting from an RB with the lowest index and ending at an RB with the highest index among RBs of the CORESET #0, and the SCS and the cyclic prefix for PDCCH reception in the CORESET #0. The UE 102 and the base station 160 may determine that frequency position of a starting (first) RB of the initial DL BWP is the frequency position C and the frequency bandwidth of the initial DL BWP is the total number C. In other words, the UE 102 and the base station 160 may determine an index of the lowest RB (starting RB, first RB) of the initial DL BWP, $N_{BWP,i}^{start, \mu}$, as the lowest RB index C, and determine the number of consecutive RBs of the initial DL BWP, $N_{BWP,i}^{size, \mu}$, is the total number C. $N_{BWP,i}^{start, \mu} (N_{BWP,i}^{start})$ herein is the common resource block where the initial DL BWP starts relative to the CRB0. $N_{BWP,i}^{size, \mu} (N_{BWP,i}^{size})$ is the frequency bandwidth of the initial DL BWP in units of RBs. For convenience of illustrations, a (initial) DL BWP defined by a location and number of contiguous PRBs, starting from an RB with the lowest index and ending at an RB with the highest index among RBs of the CORESET #0, and the SCS and the cyclic prefix for PDCCH reception in the CORESET #0 can be referred to as a DL BWP C (or an initial DL BWP C). The UE monitors, in the initial DL BWP C, PDCCH for the Type 0-PDCCH CSS set in the CORESET #0 and receive the PDSCH carrying the SIB1 according to the SCS and CP length as above-mentioned.

Upon and after the UE 102 has been provided the initial DL BWP configuration A from the SIB1, the UE 102 and the base station 160 may determine an initial DL BWP for the UE 102.

The UE 102 and the base station 160 may determine 803 whether the total number A exceeds a predefined size or not. The predefined size may refer to a maximum bandwidth the RedCap UE can support in units of RBs. Additionally or alternatively, the predefined size may be indicated by a RRC parameter included in the system information. Additionally or alternatively, the predefined size may be specified in the 3GPP specifications in advance.

In the present disclosure, the initial DL BWP configuration A may provide a configuration for a common CORESET A, i.e., the above-mentioned ControlResourceSet. Different from the CORESET #0 which is a common CORESET with index 0 provided by the MIB, the common CORESET A is a common CORESET with an index other than A and is provided by the SIB1. One or more CSS sets in the initial DL BWP configuration A can be associated with

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the common CORESET A. The UE 102 may monitor PDCCH for the one or more CSS sets in the associated common CORESET A.

The UE 102 may need to determine the frequency location and bandwidth of the common CORESET A according to a bitmap indicated by the RRC parameter frequencyDomainResource. As above-mentioned, the bits of the bitmap have a one-to-one mapping with non-overlapping groups of 6 consecutive RBs, in ascending order of RB index in a DL BWP. That is, each bit in the bitmap corresponds to a group of 6 consecutive RBs, with grouping starting from the first RB group in a DL BWP.

Therefore, in order to determine the frequency location and bandwidth of the common CORESET A, the UE 102 may need to first determine the initial DL BWP.

In the implementation 800, an example B is a case where the total number A is equal to or smaller than the predefined size. Then in the example B, the UE 102 may proceed to the operation 804, that is, the UE 102 determines the initial DL BWP of the UE 102 is the DL BWP A. In other words, the UE 102 and the base station 160 may determine an index of the lowest RB of the initial DL BWP, $N_{BWP,i}^{start, \mu}$ ($N_{BWP,i}^{start}$), as the starting RB index A, and determine the number of consecutive RBs of the initial DL BWP, $N_{BWP,i}^{size, \mu}$ ($N_{BWP,i}^{size}$), is the total number A. Therefore, $N_{BWP,i}^{start}$ of the initial DL BWP is a common resource block where the DL BWP A starts relative to the CRB0. $N_{BWP,i}^{size}$ of the initial DL BWP is the bandwidth of the DL BWP A in units of the RB.

In the example B, the UE 102 may determine the frequency location and bandwidth of the common CORESET A. For determining the frequency location of the common CORESET A, the UE 102 may need to determine a common RB index of a first common RB of the first RB group. The UE 102 may determine a common RB index of a first common RB of the first RB group by using the starting RB index A. Specifically, the UE 102 may calculate a common RB index of a first common RB of the first RB group according to $6 \times \text{ceiling}(N_{BWP,i}^{start}/6)$ wherein the $N_{BWP,i}^{start}$ is a (starting) RB index of a common RB where the initial DL BWP (i.e. the DL BWP A) starts relative to the CRB 0.

In the implementation 800, an example C is a case where the total number A is larger than the predefined size. Then in the example C, the UE 102 may proceed to the operation 805, that is, the UE 102 determines the initial DL BWP of the UE 102 is the DL BWP C. In other words, the UE 102 and the base station 160 may determine an index of the lowest RB of the initial DL BWP, $N_{BWP,i}^{start, \mu}$ ($N_{BWP,i}^{start}$), as the starting RB index C, and determine the number of consecutive RBs of the initial DL BWP, $N_{BWP,i}^{size, \mu}$ ($N_{BWP,i}^{size}$), is the total number C. Therefore, $N_{BWP,i}^{start}$ of the initial DL BWP is a common resource block where the initial DL BWP (i.e. the DL BWP C) starts relative to the CRB0. $N_{BWP,i}^{size}$ of the initial DL BWP is the bandwidth of the DL BWP C in units of the RB.

In the example C, the UE 102 may determine the frequency location and bandwidth of the common CORESET A. The UE 102 may determine a common RB index of the first common RB of the first RB group by using the starting RB index A. Specifically, the UE 102 may calculate a common RB index of a first common RB of the first RB group according to $6 \times \text{ceiling}(N_{BWP,i}^{start}/6)$ wherein the $N_{BWP,i}^{start}$ is a starting RB index of a common RB where the DL BWP A starts relative to the CRB 0. In other words, the $N_{BWP,i}^{start}$ is not a starting RB index of a common RB where the DL BWP C (i.e. the CORESET #0) starts relative to the CRB 0.

In the example C, although the UE 102 determines frequency position and bandwidth for the initial DL BWP as frequency position C and the total number C, the UE 102 may continue to use some part or all of information provided in the initial DL BWP configuration A. For example, in the initial DL BWP configuration A, the generic parameters (e.g. locationAndBandwidth, subcarrierSpacing, cyclicPrefix), the cell specific parameters (e.g. pdcch-ConfigCommon) for PDCCH, and the cell specific parameters (e.g. pdsch-ConfigCommon) for the PDSCH can be used by the UE 102 for PDCCH monitoring and PDSCH reception. For example, the UE 102 may determine SCS for the DL BWP C according to the subcarrierSpacing provided by the initial DL BWP configuration A. In other words, in the example C, the UE 102 may use the initial DL BWP configuration A for communication with the base station 160 (e.g., for PDCCH monitoring and PDSCH reception) except that the UE 102 may not use the initial DL BWP configuration A to determine the frequency position and frequency bandwidth of the initial DL BWP. On the other hand, in the example B, the UE 102 may use the initial DL BWP configuration A for communication with the base station 160 (e.g., for PDCCH monitoring and PDSCH reception) and the UE 102 also use the initial DL BWP configuration A to determine the frequency position and frequency bandwidth of the initial DL BWP.

According to the implementation 800, the RedCap UEs and the non-RedCap UEs determine different DL BWPs as their initial DL BWP when the RedCap UEs and non-RedCap UEs have been provided the initial DL BWP configuration A. The non-RedCap UEs may not determine whether the total number A exceeds the predefined size or not. The non-RedCap UEs determine the DL BWP A as the initial DL BWP. On the other hand, the RedCap UEs may not always determine the DL BWP A as the initial DL BWP. RedCap UEs may first determine whether the total number A exceeds the predefined size or not. Then the RedCap UEs may determine which one of the DL BWP A and the DL BWP C as the initial DL BWP based on whether the total number A exceeds the predefined size. The RedCap UEs may determine the DL BWP A as the initial DL BWP in a case that the total number A does not exceed the predefined size. The RedCap UEs may determine the DL BWP C as the initial DL BWP in a case that the total number A exceeds the predefined size.

For determination of a common RB index of a first common RB of the first RB group for the common CORESET A, Non-Redcap UEs may determine a common RB index of the first common RB of the first RB group by using the starting RB index of the initial DL BWP. However, the RedCap UEs may not always determine a common RB index of the first common RB of the first RB group by using the starting RB index of the initial DL BWP. The RedCap UEs may determine a common RB index of the first common RB of the first RB group by using the starting RB index A, regardless of whether the initial DL BWP is the DL BWP C or the DL BWP A.

According to the implementation, the common CORESET A can be shared between the RedCap UEs and non-RedCap UEs. Meanwhile, the base station is capable of configuring different initial DL BWPs to the RedCap UEs and non-RedCap UEs but is not at the cost of increasing the signaling overhead. Therefore, a more efficient and flexible communication can be provided by base station to UEs.

FIG. 7 is a diagram illustrating one example of initial DL BWP determination according to initial DL BWP configuration A by a UE 102 and a base station 160. The

example 700 can be applied to the various implementations and examples of the present disclosure.

FIG. 7 illustrates an example of system information (e.g., SIB1) providing one initial DL BWP configurations, i.e., the initial DL BWP configuration A and not providing the initial DL BWP configuration B. The UE 102 may determine, from the MIB, frequency position of a CORESET for Type0-PDCCH CSS set, i.e., CORESET #0 701 and the Type0-PDCCH CSS set. In the FIG. 7, 702 means the starting (first) RB of the CORESET #0 701. As illustration in the implementation 800, the UE 102 may determine the frequency position C of the starting (first) RB 702 of the CORESET #0.

The UE 102 monitors PDCCH for the Type 0-PDCCH CSS set in the CORESET #0 to receive 703 the SIB1. The base station 160 may generate SIB1 to include the initial DL BWP configuration A and to not include the initial DL BWP configuration B and broadcast/transmit to the UE 102 in a PDSCH. The PDSCH carrying SIB1 is scheduled by the PDCCH for the Type 0-PDCCH CSS set in the CORESET #0.

Before the UE 102 receives the SIB1, i.e., if the UE 102 is not provided the initial DL BWP configuration A, the UE 102 may determine that the initial DL BWP of the UE 102 is defined by a location and number of contiguous PRBs, starting from a PRB with the lowest index and ending at a PRB with the highest index among PRBs of the CORESET #0, and a SCS and a cyclic prefix for PDCCH reception in the CORESET #0. In the FIG. 7, 704 means a time duration before the UE is provided the initial DL BWP configuration A. During the time duration 704, the above-mentioned DL BWP C is the initial DL BWP for the UE 102. Likewise, during the time duration 704, the initial DL BWP (the DL BWP C) is an active DL BWP for the UE 102. The UE 102 may monitor, in the active DL BWP, PDCCH for the Type0-PDCCH CSS set and receive a PDSCH scheduled by the PDCCH for SIB1 according to the corresponding SCS and CP length as above-mentioned.

According to the received SIB1, the UE 102 may be provided, from the received SIB1, the initial DL BWP configuration A. The initial DL BWP configuration A provides the DL BWP A 705. The (total) number A of contiguous RBs, L_{RB} , of the DL BWP A is the frequency bandwidth 706 of the DL BWP A 705. An RB 707 is the starting (first) RB of the DL BWP A 705. The starting RB index (N_{BWP}^{start}) represents the RB index of a first RB 707 of the DL BWP A relative to the CRB0. As illustrated above, the UE 102 may determine, from the initial DL BWP configuration A, the frequency bandwidth of the DL BWP A and frequency position of a first RB of the DL BWP A, which are the total number A and the starting RB index A, respectively.

As depicted in the FIG. 7, the DL BWP A 705 contains the entire CORESET #0 702 in the frequency domain. In the FIG. 7, 708 means a time duration upon and/or after the UE is provided the initial DL BWP configuration A. That is, 708 means a time duration when the UE is provided the initial DL BWP configuration. The UE 102 may determine whether the frequency bandwidth exceeds the above-mentioned predefined size. That is, the UE 102 may determine whether the total number A exceeds the predefined size or not. And the UE 102 may determine, during the time duration 708, which of the DL BWP A and the DL BWP C as the initial DL BWP based on whether the total number A exceeds the predefined size.

In a case that the frequency bandwidth of the DL BWP A exceeds the predefined size, the UE 102 may determine, during the time duration 708, the DL BWP C is the initial DL BWP of the UE 102. During the time duration 708, the DL

BWP C continue to be determined as the initial DL BWP for the UE 102. Likewise, during the time duration 708, the DL BWP C is the active DL BWP for the UE 102. That is, during the time duration 708, the bandwidth of the initial DL BWP is the frequency bandwidth 709 of the DL BWP C which is the total number C. Likewise, during the time duration 708, the first RB of the initial DL BWP is the first RB 702 of the DL BWP C which corresponds to the frequency position C (the starting RB index C). During the time duration 708, the UE 102 may monitor, in the active DL BWP, PDCCH for the common CSS sets (e.g., Type0A-PDCCH CSS set, the Type1-PDCCH CSS set, and/or the Type2-PDCCH CSS set) and receive a PDSCH scheduled by the PDCCH according to the SCS and CP length as determined for the DL BWP C. In this case, the UE 102 may not switch the active DL BWP and continue to determine the active DL BWP (initial DL BWP) is the DL BWP C even if the UE 102 has been provided the initial DL BWP configuration A. That is, for the UE 102, the initial DL BWP during the time duration 704 and the initial DL BWP during the time duration 708 are corresponding to a same DL BWP.

In a case that the frequency bandwidth of the DL BWP A does not exceed the predefined size, the UE 102 may determine, during the time duration 708, the DL BWP A is the initial DL BWP of the UE 102. During the time duration 708, the DL BWP A is the initial DL BWP for the UE 102. Likewise, during the time duration 708, the DL BWP A is an active DL BWP for the UE 102. That is, during the time duration 708, the bandwidth of the initial DL BWP is the frequency bandwidth 706 of the DL BWP A which is the total number A. Likewise, during the time duration 708, the first RB of the initial DL BWP is the first RB 707 of the DL BWP A which corresponds to the starting RB index A. During the time duration 708, the UE 102 may monitor, in the active DL BWP, PDCCH for the common CSS sets (e.g., Type0A-PDCCH CSS set, the Type1-PDCCH CSS set, and/or the Type2-PDCCH CSS set) and receive a PDSCH scheduled by the PDCCH according to the SCS and CP length as determined for the DL BWP A. In this case, the UE 102 may switch the active DL BWP from the DL BWP C to the DL BWP A upon the UE 102 has been provided the initial DL BWP configuration A. That is, for the UE 102, the initial DL BWP during the time duration 704 and the initial DL BWP during the time duration 708 are corresponding to different DL BWPs.

The initial DL BWP configuration A may include a RRC parameter ControlResourceSet to indicate a CORESET (a common CORESET A). One or more CSS sets in the initial DL BWP configuration A can be associated with the common CORESET A. Here, the CORESET indicated by the RRC parameter ControlResourceSet in the initial DL BWP configuration A can be regarded as the common CORESET A 710.

The UE 102 may apply a RRC parameter frequencyDomainResource indicating a bitmap with 45 bits to determine the frequency position of the common CORESET A 710 wherein the RRC parameter frequencyDomainResource is included in the RRC parameter ControlResourceSet provided by the initial DL BWP configuration A. Each bit in the bitmap corresponds an RB group of 6 consecutive RBs, with grouping starting from the first RB group in the DL BWP A 706. The first (left-most/most significant) bit of the bitmap corresponds to the first RB group. To determine the frequency position of the common CORESET A, the UE 102 may first determine the frequency position of the first RB group.

In the case that the frequency bandwidth of the DL BWP A exceeds the predefined size, the UE 102 may determine, during the time duration 708, the DL BWP C is the initial DL BWP of the UE 102. Although the DL BWP C is the initial DL BWP during the time duration 708, the UE may determine a common RB index of the first common RB of the first RB group by using the starting RB index A.

As depicted in the FIG. 7, RB group 711 is the first RB group corresponding to the first bit of the bitmap. The RB group 712 is the fifth RB group corresponding to the fifth bit of the bitmap. Here, the first bit of the bitmap is set to 0, which means the RBs of the first RB group do not belong to the frequency domain resource of the common CORESET A. On the other hand, the fifth bit of the bitmap is set to 1, which means the RBs of the fifth RB group 712 belong to the frequency domain resource of the common CORESET A.

Additionally or alternatively, the base station 160 may generate the system information (e.g., SIB1) to include the initial DL BWP configuration A and to not include the initial DL BWP configuration B. In a case that the UE 102 is provided the initial DL BWP configuration A and the total number A exceeds the predefined size, the UE 102 may determine the DL BWP A as the initial DL BWP, i.e. the UE 102 may determine the total number A and the starting RB index A as the frequency bandwidth of the initial DL BWP and the frequency position (RB index) of a first RB of the initial DL BWP, respectively. If the UE has been provided the initial DL BWP configuration A, the UE 102 may switch the active DL BWP from the DL BWP C to the DL BWP A when the UE enters into the during the time duration 708 from the time duration 704. That is, the DL BWP A is the active DL BWP for the UE 102. Then, the UE may perform reception of PDCCH and PDSCH in the active DL BWP according to the SCS and CP length determined for the DL BWP A.

Upon successful completion of the random access procedure, the UE 102 may enter into an RRC_CONNECTED state. The UE 102 may move from either RRC_IDLE state or RRC_INACTIVE to the RRC_CONNECTED state. The UE 102 in the RRC_CONNECTED state may be provided a C-RNTI by the base station 160. In a case that the UE 102 is provided the C-RNTI, the UE 102 may determine the DL BWP C as the initial DL BWP, i.e. the UE 102 may determine the total number C and the starting RB index C as the frequency bandwidth of the initial DL BWP and the frequency position (RB index) of a first RB of the initial DL BWP, respectively. That is, in a case that the UE 102 is provided the C-RNTI, the UE 102 may switch the active DL BWP from the DL BWP A to the DL BWP C. That is, the DL BWP C is the active DL BWP for the UE 102. In other words, when the UE is in the RRC_CONNECTED state and the active DL BWP is the DL BWP C, the UE 102 may perform reception of PDCCH and PDSCH in the DL BWP C according to the SCS and CP length determined for the DL BWP C.

In another implementation of the present disclosure, the base station 160 may generate system information (e.g., SIB1) to include one or both of the initial DL BWP configuration A and the initial DL BWP configuration B. The UE 102 may initiate a random access procedure on a serving cell (e.g., a PCell). In a case that PRACH occasions are not configured for the active UL BWP of the serving cell, the UE 102 may switch the active UL BWP to an initial UL BWP and may switch the active DL BWP to which DL BWP based on the initial DL BWP configuration A and/or the initial DL BWP configuration B. That is, the UE 102 may determine whether the system information includes the initial DL BWP

configuration B or not and determine to switch the active DL BWP to which DL BWP based on whether the system information includes the initial DL BWP configuration B or not. In a case that the system information includes the initial DL BWP configuration B, the UE 102 may switch the active DL BWP to the DL BWP B indicated by the initial DL BWP configuration B.

In a case that the system information does not include the initial DL BWP configuration B, the UE 102 may switch the active DL BWP to a DL BWP between DL BWP A and DL BWP C based on whether the total number A exceeds the predefined size or not. That is, in this case, the UE 102 may determine whether the total number A exceeds the predefined size or not and then determine to switch the active DL BWP to which of the DL BWP A and DL BWP C based on whether the total number A exceeds the predefined size or not. In a case that the total number A exceeds the predefined size, the UE 102 may switch the active DL BWP to DL BWP C. In a case that the total number A does not exceed the predefined size, the UE 102 may switch the active DL BWP to the DL BWP A indicated by the initial DL BWP configuration A. After switching, the UE perform the random access procedure on the active DL BWP and the active UL BWP of the serving cell.

As above-mentioned, the base station may configure one or two initial DL BWP configurations for a serving cell. The UE 102 may camp on the serving cell and may receive the system information including the one or two initial DL BWP configurations. Before the UE 102 receives the system information which includes the initial DL BWP configuration A and/or the initial DL BWP configuration B, i.e., the UE 102 is not provided the initial DL BWP configuration A and/or the initial DL BWP configuration B, the UE 102 may determine that an initial DL BWP is defined by a location and number of contiguous PRBs, starting from a PRB with the lowest index and ending at a PRB with the highest index among PRBs of the CORESET #0, and a SCS and a cyclic prefix for PDCCH reception in the CORESET #0. If the UE 102 receives the system information which includes the initial DL BWP configuration A and does not include the initial DL BWP configuration B for a serving cell, i.e., the UE 102 is provided the initial DL BWP configuration A and is not provided the initial DL BWP configuration B, the UE 102 may determine, based on whether the total number A exceeds the predefined size or not, that an initial DL BWP is a DL BWP provided by the initial DL BWP configuration A or a DL BWP defined by a location and number of contiguous PRBs, starting from a PRB with the lowest index and ending at a PRB with the highest index among PRBs of the CORESET #0, and a SCS and a cyclic prefix for PDCCH reception in the CORESET #0.

In another implementation of the present disclosure, the base station 160 may generate the system information (e.g., SIB1) to include the initial DL BWP configuration A and to not include the initial DL BWP configuration B. In a case that the UE 102 is provided the initial DL BWP configuration A and the total number A exceeds the predefined size, the UE 102 may determine the DL BWP A as the initial DL BWP during the time duration 708. However, the base station 160 would schedule the reception of PDCCH and PDSCH to be configured within the frequency bandwidth of the CORESET #0 such that the UE 102 can successfully receive the PDCCH and the corresponding PDSCH. The base station 160 may transmit, in a USS set, a PDCCH with a DCI format to the UE 102 in an active DL BWP. Here, the DL BWP A is the active DL BWP. The DCI format is used to schedule a PDSCH transmission. The base station 160

may generate a frequency domain resource allocation (FDRA) field in the DCI format wherein the bitwidth (size) of FDRA field is determined or calculated by using the size of the CORESET #0 (i.e., the total number C). That is, the base station 160 may not use the size of the DL BWP A (i.e., the total number A) to determine the size of the FDRA field. Likewise, the UE 102 may determine the bitwidth of the FDRA field by using the size of the CORESET #0 (i.e., the total number C). That is, the UE 102 may not use the size of the DL BWP A to determine or calculate the bitwidth of the FDRA field in the DCI format.

On the other hand, the base station 160 may generate the system information (e.g., SIB1) to include the initial DL BWP configuration A and to not include the initial DL BWP configuration B. In a case that the UE 102 is provided the initial DL BWP configuration A and the total number A does not exceed the predefined size, the UE 102 may determine the DL BWP A as the initial DL BWP during the time duration 708. The base station 160 may transmit, in a USS set, a PDCCH with a DCI format to the UE 102 in an active DL BWP. Here, the DL BWP A is the active DL BWP. The DCI format is used to schedule a PDSCH transmission. The base station 160 may generate a frequency domain resource allocation (FDRA) field in the DCI format wherein the bitwidth (size) of FDRA field is determined or calculated by using the size of the active DL BWP (i.e., the size of the DL BWP A). Likewise, the UE 102 may determine the bitwidth of the FDRA field by using the size of the active DL BWP (i.e., the size of the DL BWP A). That is, the UE 102 may not use the size of the DL BWP A to determine or calculate the bitwidth of the FDRA field in the DCI format.

In various implementations of the present disclosure, for the UE 102, the DL BWP C, the DL BWP A and the DL BWP B cannot be active simultaneously. That is, only one of the DL BWP C, the DL BWP A and the DL BWP B would be activated by the UE 102 at a given time. To ensure an efficient communication between the base station and the UE 102, the UE 102 and/or the base station may need to determine which DL BWP between the DL BWP A and the DL BWP B is determined as an initial DL BWP for reception of PDCCH and/or PDSCH for one or more CSS sets. A CSS set herein may refer to a Type0-PDCCH CSS set, a Type0A-PDCCH CSS set, a Type1-PDCCH CSS set, or a Type2-PDCCH CSS set. In the present disclosure, the UE 102 needs to monitor PDCCH for the Type0-PDCCH CSS set to receive SIB1 message, the UE 102 needs to monitor PDCCH for the Type0A-PDCCH CSS set to receive system information other than MIB and SIB1 message, the UE 102 needs to monitor PDCCH for the Type1-PDCCH CSS set for random access procedure, and, the UE 102 needs to monitor PDCCH for the Type2-PDCCH CSS set to receive paging message. When the UE 102 monitors PDCCH for a CSS set on a DL BWP, the DL BWP where the UE 102 monitors PDCCH for the CSS set is an active DL BWP to the UE 102. The active DL BWP where the UE 102 monitors PDCCH for a CSS set may be different from the active DL BWP where the UE 102 monitors PDCCH for another CSS set.

In the present disclosure, the base station 160 may not configure the UE 102 without one CSS set on the DL BWP A provided by the initial DL BWP configuration A. Similarly, the base station 160 may not configure the UE 102 with the initial DL BWP configuration B, at least one CSS set would be included in the initial DL BWP configuration B. That is, the base station 160 may configure at least one CSS set on the DL BWP A. A CSS set herein may refer to a Type0-PDCCH CSS set, a Type0A-PDCCH CSS set, a Type1-PDCCH CSS set, or a Type2-PDCCH CSS set. For

one initial DL BWP configuration, a RRC parameter searchSpaceSIB1 is used to indicate the Type0-PDCCH CSS set for SIB1 message, a RRC parameter searchSpaceOther-SystemInformation is used to indicate the Type0A-PDCCH CSS set for other system information, a RRC parameter pagingSearchSpace is used to indicate the Type2-PDCCH CSS set for paging, and a RRC parameter ra-SearchSpace is used to indicate the Type1-PDCCH CSS set for random access procedure.

In various implementations or examples of the present disclosure, as long as the UE 102 and/or the base station 160 determine which of the DL BWP A, the DL BWP C, and the DL BWP B is the initial DL BWP, for example, the UE 102 may monitor PDCCH and receive PDSCH according to a SCS and a CP length configured for the DL BWP which is determined as the initial DL BWP. Additionally, the UE 102 may apply the frequency location and/or bandwidth of the DL BWP (i.e., the starting position $N_{BWP,i}^{start, \mu}$ of the DL BWP), which is determined as the initial DL BWP, to determine mapping from virtual RB to physical RB for PDSCH schedule by the CSS set.

According to the above implementations of the present disclosure, a more efficient and flexible communication can be provided by base station to UEs. The base station can determine whether to share control signaling and share which control signaling to RedCap UEs and non-RedCap UEs. According to determination of initial DL BWP among the DL BWP A, the DL BWP B and the DL BWP C, the UE 102 can be aware of how and where to monitor PDCCH for a CSS set and to receive a corresponding PDSCH in different phases in which DL BWP.

Hereinafter, random access procedure is described.

The random access procedure is used to transmit the RRCSetupRequest message, the RRCResumeRequest message, and/or the RRCReestablishmentRequest message. The RRCSetupRequest message is used to request the establishment of an RRC connection between the UE 102 and the network (the base station 160). The RRCResumeRequest message is used to request the resumption of a suspended RRC connection or perform an RNA update. The RRCReestablishmentRequest message is used to request the reestablishment of an RRC connection.

In the present disclosure, two types of random access procedure are supported, i.e. 4-step random access procedure and 2-step random access procedure. The 4-step random access procedure can be also referred to as Type-1 random access procedure or as 4-step random access type. The 2-step random access procedure can be also referred to as Type-2 random access procedure or as 2-step random access type. Both types of the random access procedure support contention-based random access (CBRA) and contention-free random access (CFRA).

4-step Random access procedure may include the transmission of random access preamble (Msg1 or Message 1) in a PRACH, the reception of random access response (RAR) message with a PDCCH and/or a PDSCH (Msg2, Message 2), and when applicable, the transmission of a PUSCH scheduled by a RAR UL grant (e.g., Msg 3, Message 3), and the reception of PDSCH for contention resolution.

2-step Random access procedure may include the transmission of random access preamble in a PRACH and of a PUSCH (MsgA), and the reception of a RAR message with a PDCCH and/or a PDSCH (MsgB), and when applicable, the transmission of a PUSCH scheduled by a fallback RAR UL grant, and the reception of PDSCH for contention resolution.

The base station may transmit a set of SS/PBCH blocks in a serving cell and indicate the indices of the transmitted SS/PBCH blocks within a half-frame to UEs camping on the serving cell via SIB1. In other words, the base station 160 may indicate the time domain positions of the transmitted SS/PBCH blocks within a half frame. As above-mentioned, upon detection of a SS/PBCH block with an index, a UE may determine from the MIB a CORESET for Type0-PDCCH CSS set and the Type0-PDCCH CSS set. The UE monitors PDCCH in the Type 0-PDCCH CSS set to receive the SIB1. Then according to the received SIB1, the UE may determine, within a half-frame, a set of SS/PBCH blocks which are transmitted by the base station. In other words, the UE may determine, within a half-frame, the time domain positions of a set of SS/PBCH blocks which are transmitted by the base station.

Before initiating a random access procedure, the UE 102 may, based on the received SIB1, obtain a set of SS/PBCH block indexes. A set of SS/PBCH blocks corresponding to the indexes in the set of SS/PBCH block indexes are transmitted by the base station. That is, the base station 160 may notify the UE 102 of the set of SS/PBCH blocks that are transmitted by the base station via a parameter included in the SIB1. The UE 102 may perform reference signal received power (RSRP) measurements for the set of SS/PBCH blocks. On the other hand, the UE 102 may not perform RSRP measurements for those candidate SS/PBCH blocks which are not transmitted by the base station.

The secondary synchronization signals of a SS/PBCH block is used for the RSRP determination for the corresponding SS/PBCH block. The number of resource elements carrying the secondary synchronization signals of the SS/PBCH block (or the SS/PBCH blocks with the same SS/PBCH block index) within a measurement period may be used by the UE 102 to determine the RSRP of the SS/PBCH block. Additionally, the demodulation reference signals for PBCH of the SS/PBCH block and/or configured CSI reference signals can also be used by the UE 102 to determine the RSRP of the SS/PBCH block.

Before initiating a random access procedure, the UE 102 may receive, from the base station 160, the information for the random access procedure. The information (i.e. cell specific random access configuration(s)) includes the cell specific random access parameters and/or the dedicated random access parameters. The random access information may be indicated by the broadcasted system information (e.g., MIB, SIB1, and/or other SIBs) and/or RRC message and so on. For example, the information may include the configuration of PRACH transmission parameters such as time resources for PRACH transmission, frequency resources for PRACH transmission, the PRACH preamble format, preamble SCS and so on. The information may also include parameters for determining the root sequences (logical root sequence index, root index) and their cyclic shifts (CSs) in the PRACH preamble sequence set.

The random access preamble (PRACH preamble, or preamble) sequence is based on the Zadoff-Chu sequence. The logical root for the Zadoff-Chu sequence is provided by the information as above-mentioned. That is, a UE can generate a set of PRACH preamble sequences based on the Zadoff-Chu sequence corresponding to a root sequence indicated by the base station 160. There are two sequence lengths for the preamble. One is 839 and the other one is 139.

A preamble is transmitted by the UE 102 in a time-frequency PRACH occasion. A PRACH occasion is a time-frequency resource where the base station configures to multiple UEs for preamble transmission. There are 64 pre-

ambles defined in each time-frequency PRACH occasion. In other words, the UE 102 may generate 64 preambles for each PRACH occasion. The preambles (e.g. 64 preambles) in one PRACH occasion may be generated by one root Zadoff-Chu sequence or more than one root Zadoff-Chu sequences. The number of preambles generated from a single root Zadoff-Chu sequence at least depends on the sequence length and/or a distance of the cyclic shifts between two preambles with consecutive preamble indices. The distance of the cyclic shifts is provided by the base station 160.

Therefore, in some cases, the UE 102 can generate 64 preambles from a single root Zadoff-Chu sequence. In some cases, the UE 102 cannot generate 64 preambles from a single root Zadoff-Chu sequence. In these cases, in order to obtain the 64 preambles in a PRACH occasion, the UE 102 needs to generate the 64 preambles from multiple root Zadoff-Chu sequences with multiple consecutive root indices. The starting root index of the multiple consecutive root indices is indicated by the base station 160. The UE 102 and the base station 160 may enumerate the 64 preambles in increasing order of first increasing cyclic shift (CS) of a logical root Zadoff-Chu sequence, and then in increasing order of the logical root sequence index. The preamble indices for 64 preambles in a PRACH occasion are from 0 to 63.

The random access information (i.e., random access configuration(s)) may include a RRC parameter indicating how many SS/PBCH blocks is associated with a PRACH occasion. For example, if a value indicated by the RRC parameter is one half (i.e. $\frac{1}{2}$), it implies that one SS/PBCH block is associated with two PRACH occasions. For example, if a value indicated by the RRC parameter is two (i.e. 2), it implies that two SS/PBCH blocks are associated with one PRACH occasion.

In addition, the random access information may include a RRC parameter indicating how many frequency multiplexed PRACH occasions there are in one time instance. The random access information may include a RRC parameter indicating an offset of lowest PRACH occasion in frequency domain with respect to PRB0 of the active UL BWP. The UE 102 may determine starting symbol of a PRACH occasion, a number of PRACH occasions in time domain within a PRACH slot, a duration in symbols of the PRACH occasion according to the random access information.

As above-mentioned, SIB1 indicates a set of SS/PBCH blocks which are transmitted by the base station. In other words, the SIB1 provides SS/PBCH block indexes with which a set of SS/PBCH blocks are transmitted by the base station. The base station and/or the UE may only map the SS/PBCH indexes provided in the SIB1 to the PRACH occasions in accordance with the following rules: (i) first, in increasing order of preamble indexes within a single PRACH occasion, (ii) second, in increasing order of frequency resource indexes for frequency multiplexed PRACH occasions, (iii) third, in increasing order of time resource indexes for time multiplexed PRACH occasions within a PRACH slot, (iv) in increasing order of indexes for PRACH slots.

FIG. 9 illustrates various components that may be utilized in a UE 902. The UE 902 (UE 102) described in connection with FIG. 9 may be implemented in accordance with the UE 102 described in connection with FIG. 1. The UE 902 includes a processor 981 that controls operation of the UE 902. The processor 981 may also be referred to as a central processing unit (CPU). Memory 987, which may include read-only memory (ROM), random access memory (RAM),

a combination of the two or any type of device that may store information, provides instructions 983a and data 985a to the processor 981. A portion of the memory 987 may also include non-volatile random access memory (NVRAM). Instructions 983b and data 985b may also reside in the processor 981. Instructions 983b and/or data 985b loaded into the processor 981 may also include instructions 983a and/or data 985a from memory 987 that were loaded for execution or processing by the processor 981. The instructions 983b may be executed by the processor 981 to implement one or more of the methods 200 described above.

The UE 902 may also include a housing that contains one or more transmitters 958 and one or more receivers 920 to allow transmission and reception of data. The transmitter(s) 958 and receiver(s) 920 may be combined into one or more transceivers 918. One or more antennas 922a-n are attached to the housing and electrically coupled to the transceiver 918.

The various components of the UE 902 are coupled together by a bus system 989, which may include a power bus, a control signal bus and a status signal bus, in addition to a data bus. However, for the sake of clarity, the various buses are illustrated in FIG. 9 as the bus system 989. The UE 902 may also include a digital signal processor (DSP) 991 for use in processing signals. The UE 902 may also include a communications interface 993 that provides user access to the functions of the UE 902. The UE 902 illustrated in FIG. 9 is a functional block diagram rather than a listing of specific components.

FIG. 10 illustrates various components that may be utilized in a base station 1060. The base station 1060 described in connection with FIG. 10 may be implemented in accordance with the base station 160 described in connection with FIG. 1. The base station 1060 includes a processor 1081 that controls operation of the base station 1060. The processor 1081 may also be referred to as a central processing unit (CPU). Memory 1087, which may include read-only memory (ROM), random access memory (RAM), a combination of the two or any type of device that may store information, provides instructions 1083a and data 1085a to the processor 1081. A portion of the memory 1087 may also include non-volatile random access memory (NVRAM). Instructions 1083b and data 1085b may also reside in the processor 1081. Instructions 1083b and/or data 1085b loaded into the processor 1081 may also include instructions 1083a and/or data 1085a from memory 1087 that were loaded for execution or processing by the processor 1081. The instructions 1083b may be executed by the processor 1081 to implement one or more of the methods 300 described above.

The base station 1060 may also include a housing that contains one or more transmitters 1017 and one or more receivers 1078 to allow transmission and reception of data. The transmitter(s) 1017 and receiver(s) 1078 may be combined into one or more transceivers 1076. One or more antennas 1080a-n are attached to the housing and electrically coupled to the transceiver 1076.

The various components of the base station 1060 are coupled together by a bus system 1089, which may include a power bus, a control signal bus and a status signal bus, in addition to a data bus. However, for the sake of clarity, the various buses are illustrated in FIG. 10 as the bus system 1089. The base station 1060 may also include a digital signal processor (DSP) 1091 for use in processing signals. The base station 1060 may also include a communications interface 1093 that provides user access to the functions of the

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base station 1060. The base station 1060 illustrated in FIG. 10 is a functional block diagram rather than a listing of specific components.

The term “computer-readable medium” refers to any available medium that can be accessed by a computer or a processor. The term “computer-readable medium,” as used herein, may denote a computer- and/or processor-readable medium that is non-transitory and tangible. By way of example, and not limitation, a computer-readable or processor-readable medium may comprise RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to carry or store desired program code in the form of instructions or data structures and that can be accessed by a computer or processor. Disk and disc, as used herein, includes compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray® disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers.

It should be noted that one or more of the methods described herein may be implemented in and/or performed using hardware. For example, one or more of the methods described herein may be implemented in and/or realized using circuitry, a chipset, an application-specific integrated circuit (ASIC), a large-scale integrated circuit (LSI) or integrated circuit, etc.

Each of the methods disclosed herein comprises one or more steps or actions for achieving the described method. The method steps and/or actions may be interchanged with one another and/or combined into a single step without departing from the scope of the claims. In other words, unless a specific order of steps or actions is required for proper operation of the method that is being described, the order and/or use of specific steps and/or actions may be modified without departing from the scope of the claims.

It is to be understood that the claims are not limited to the precise configuration and components illustrated above. Various modifications, changes and variations may be made in the arrangement, operation and details of the systems, methods and apparatus described herein without departing from the scope of the claims.

The invention claimed is:

1. A user equipment (UE), comprising:

reception circuitry configured to receive, from a base station, MIB providing a first quantity of contiguous resource blocks (RBs) for a first CORESET and a first frequency position of a first starting RB for the first CORESET, and to receive system information including a second configuration of an initial downlink (DL) BWP, the second configuration providing a second quantity of contiguous RBs, a second frequency position of a second starting RB, and a bitmap indicating frequency domain resources for a second CORESET, each bit of the bitmap corresponds a group of 6 contiguous RBs, with grouping starting from a first RB group in ascending order of RB index; and control circuitry configured to determine whether the second quantity exceeds a predefined size, wherein the control circuitry is further configured to,

in a case the second quantity exceeds the predefined size, determine the first frequency position and the first quantity as a starting position and a bandwidth for an initial DL BWP, and determine a first RB index of the first RB group by using the second frequency position, and

in a case the second quantity does not exceed the predefined size, determine the second frequency

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position and the second quantity as the starting position and the bandwidth, and determine the first RB index of the first RB group by using the second frequency position.

2. The UE according to the claim 1:

the predefined size corresponds to a maximum bandwidth the UE can support in units of RBs.

3. A base station, comprising:

control circuitry configured to generate, to a user equipment (UE), a MIB to provide a first quantity of contiguous resource blocks (RBs) for a first CORESET and a first frequency position of a first starting RB for the first CORESET,

to generate system information to include a second configuration of an initial downlink (DL) BWP, the second configuration providing a second quantity of contiguous RBs, a second frequency position of a second starting RB, and a bitmap indicating frequency domain resources for a second CORESET, each bit of the bitmap corresponds a group of 6 contiguous RBs, with grouping starting from a first RB group in ascending order of RB index,

to determine whether the second quantity exceeds a predefined size, wherein the control circuitry is further configured to,

in a case the second quantity exceeds the predefined size, determine the first frequency position and the first quantity as a starting position and a bandwidth for an initial DL BWP of the UE, and determine a first RB index of the first RB group by using the second frequency position, and

in a case the second quantity does not exceed the predefined size, determine the second frequency position and the second quantity as the starting position and the bandwidth, and determine the first RB index of the first RB group by using the second frequency position; and

transmission circuitry configured to transmit the MIB and the system information.

4. The base station according to the claim 1:

the predefined size corresponds to a maximum bandwidth the UE can support in units of RBs.

5. A communication method performed by a user equipment (UE), comprising:

receiving, from a base station, MIB providing a first quantity of contiguous resource blocks (RBs) for a first CORESET and a first frequency position of a first starting RB for the first CORESET;

receiving system information including a second configuration of an initial downlink (DL) BWP, the second configuration providing a second quantity of contiguous RBs, a second frequency position of a second starting RB, and a bitmap indicating frequency domain resources for a second CORESET, each bit of the bitmap corresponds a group of 6 contiguous RBs, with grouping starting from a first RB group in ascending order of RB index; and

determining whether the second quantity exceeds a predefined size, wherein

in a case the second quantity exceeds the predefined size, determining the first frequency position and the first quantity as a starting position and a bandwidth for an initial DL BWP, and determining a first RB index of the first RB group by using the second frequency position, and

in a case the second quantity does not exceed the predefined size, determining the second frequency

position and the second quantity as the starting position and the bandwidth, and determining the first RB index of the first RB group by using the second frequency position.

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